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Chen

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(54) **PHOTOGRAPHING LENS ASSEMBLY,
IMAGE CAPTURING DEVICE AND MOBILE
TERMINAL**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 51 days.

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G02B 13/18 (2006.01)

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CPC **G02B 13/0045** (2013.01)

(58) **Field of Classification Search**

CPC G02B 13/0045; H04N 1/00

USPC 348/240.3; 259/716

See application file for complete search history.

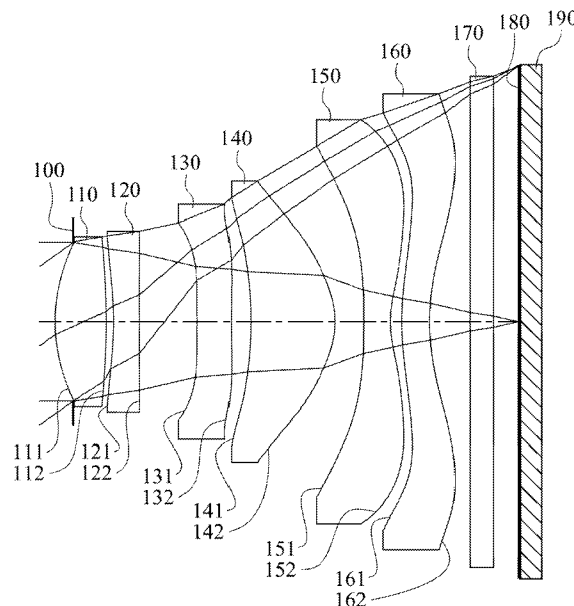
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(57) **ABSTRACT**

A photographing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element, which has a total of six non-cemented lens elements with refractive power. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element with refractive power has an object-side surface being concave in a paraxial region thereof. The third and the fourth lens elements have refractive power. The fifth lens element with negative refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof.

18 Claims, 19 Drawing Sheets



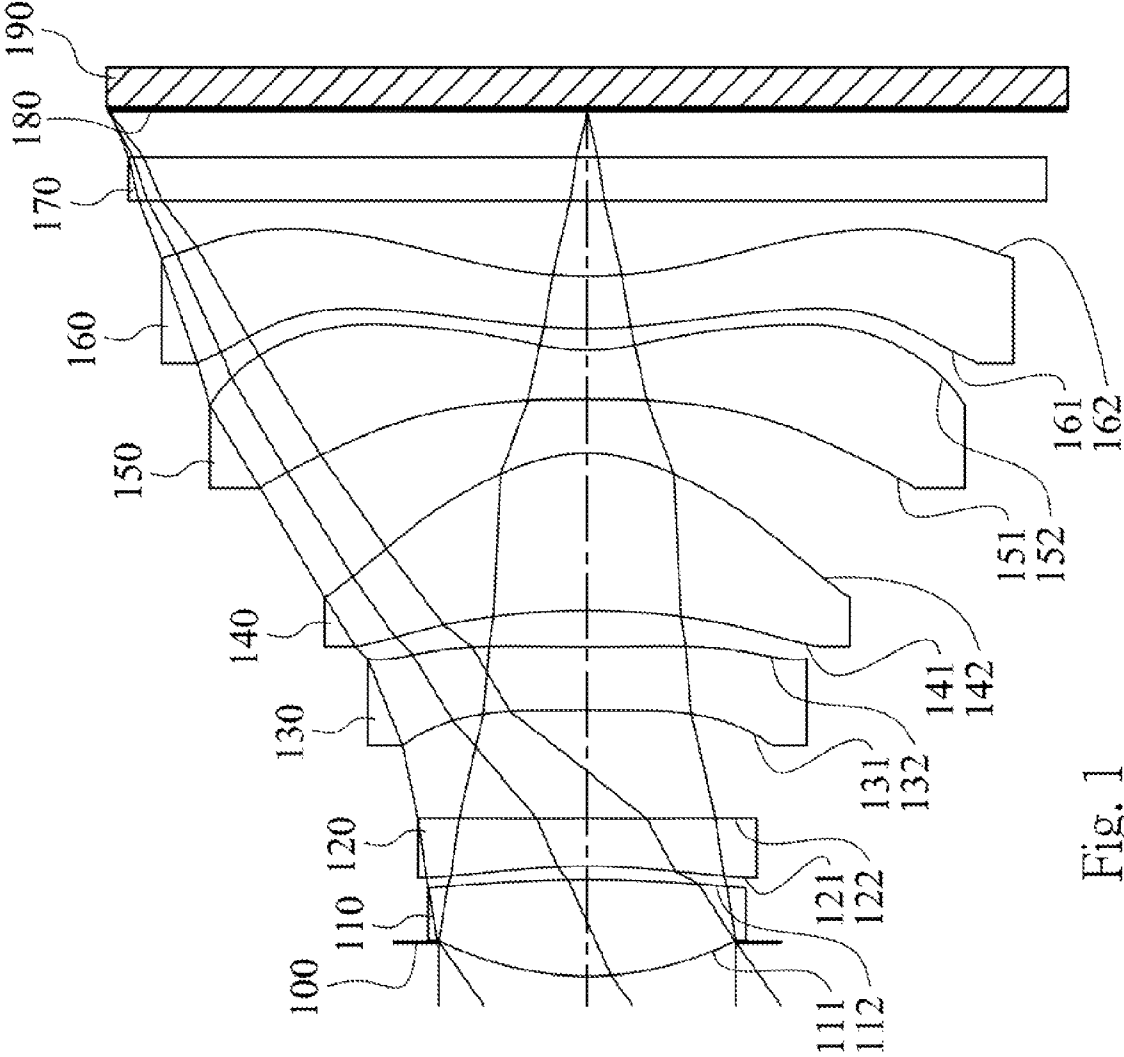


Fig. 1

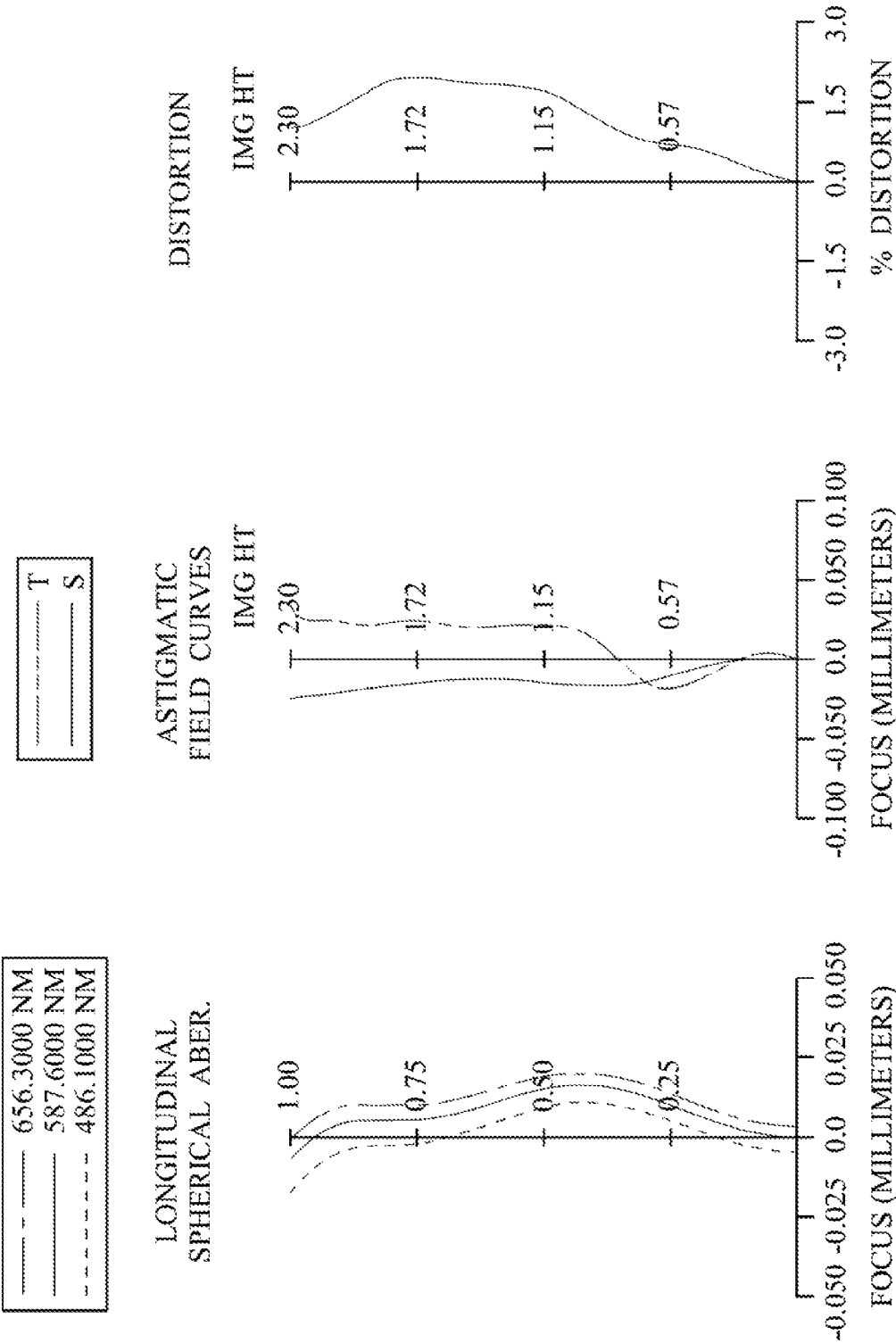


Fig. 2

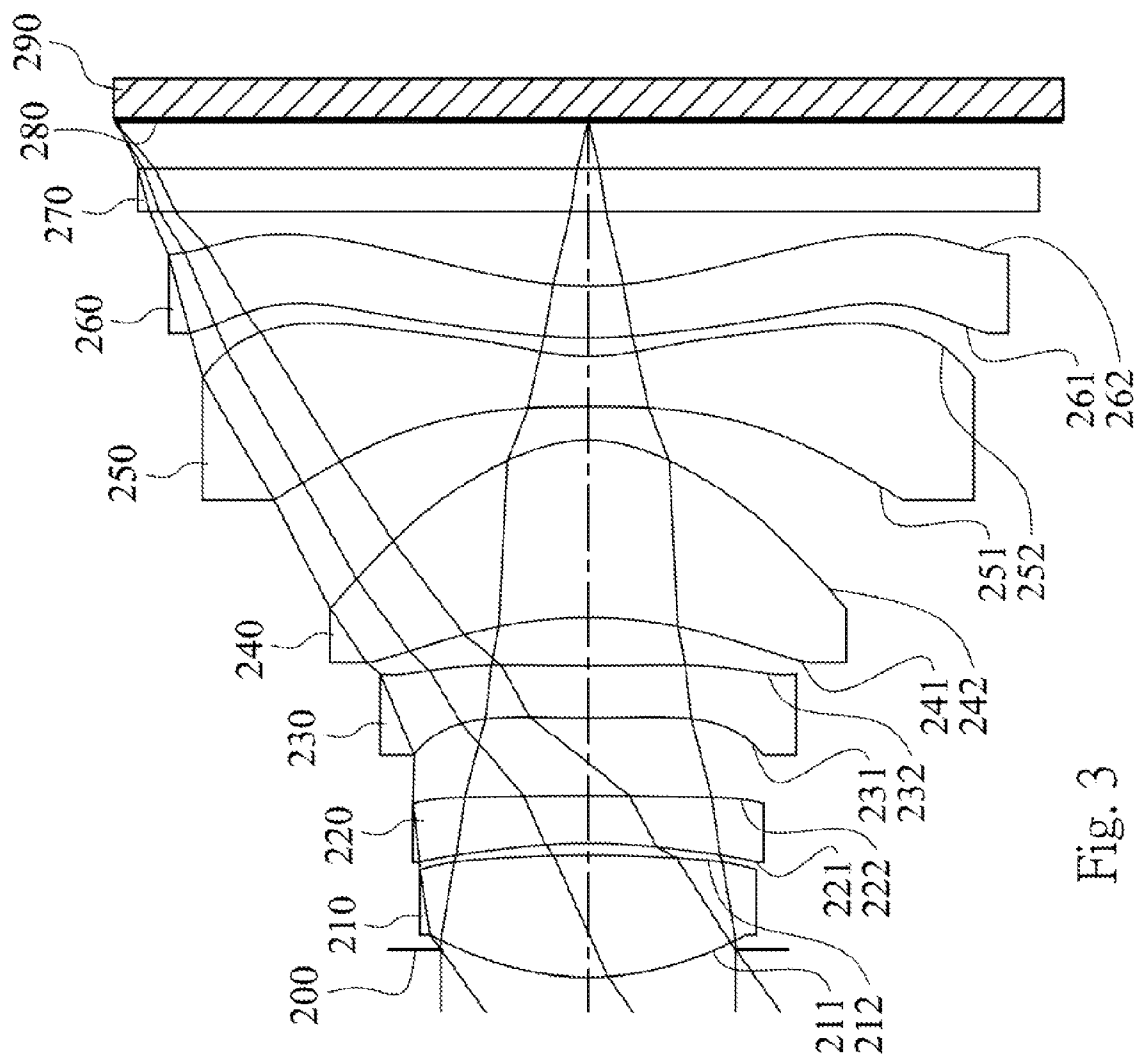


Fig. 3

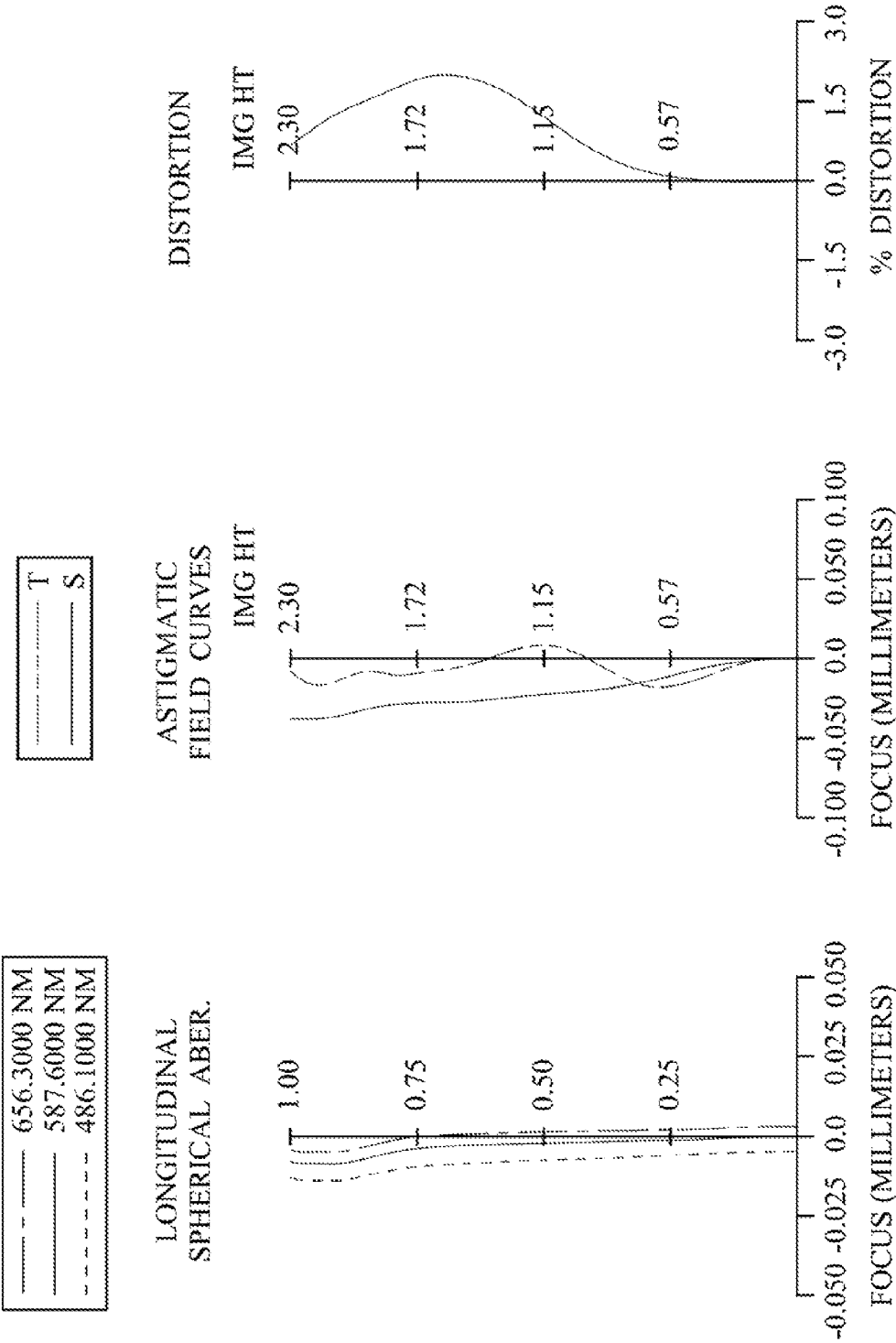


Fig. 4

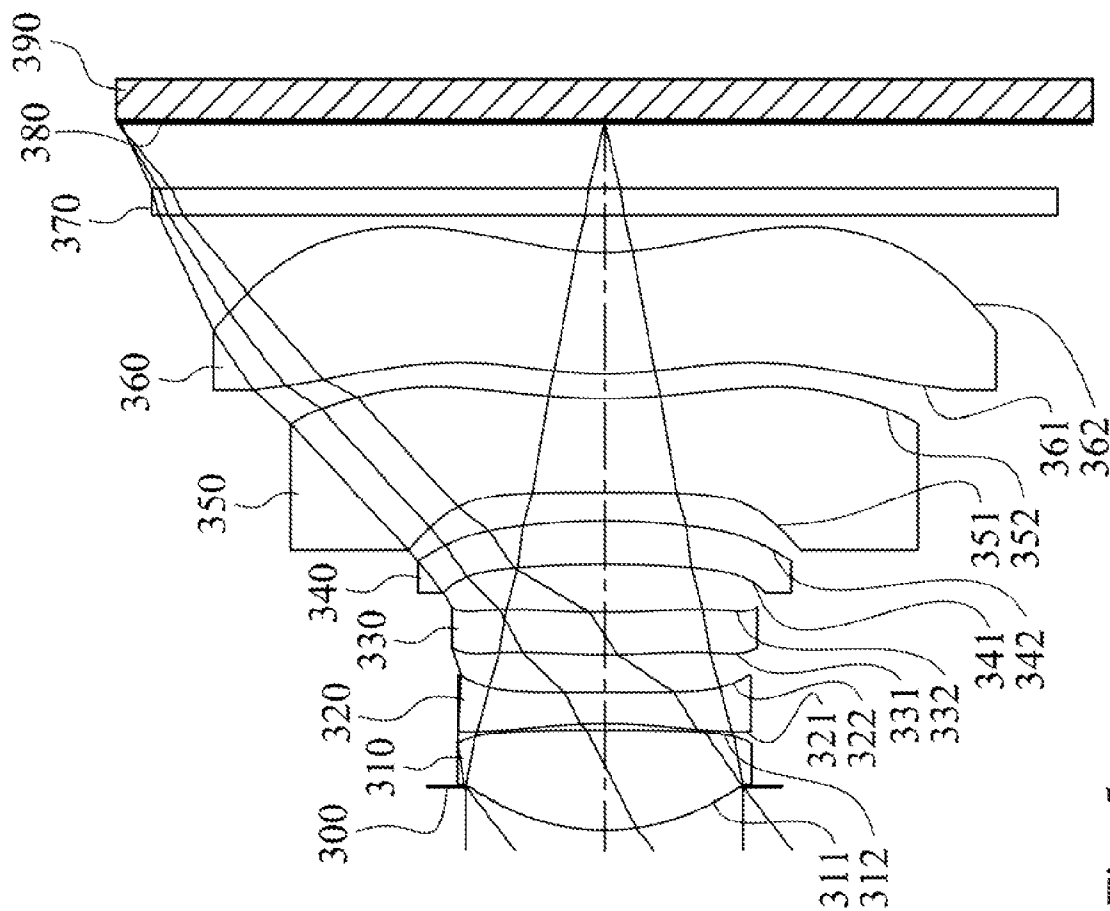


Fig. 5

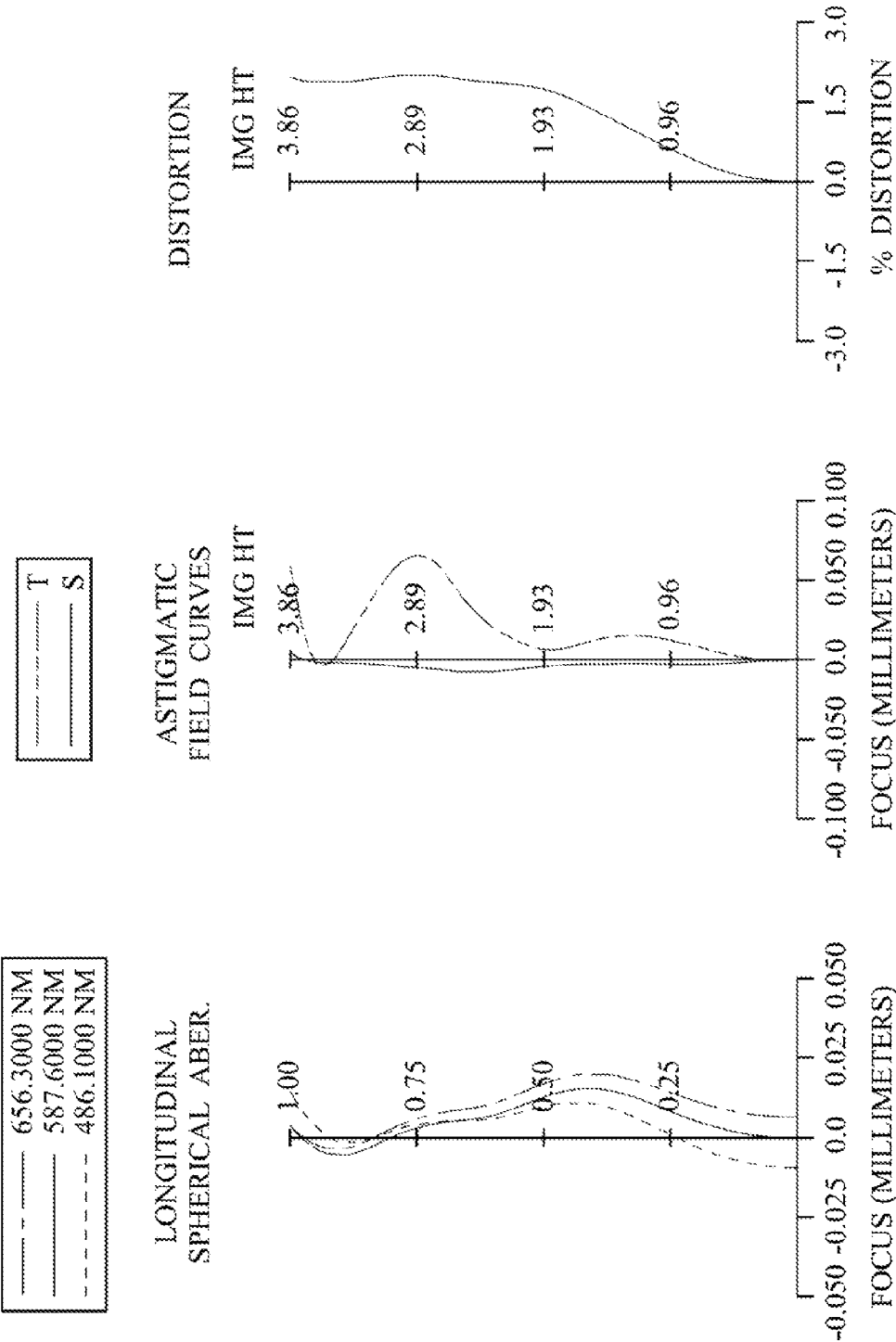


Fig. 6

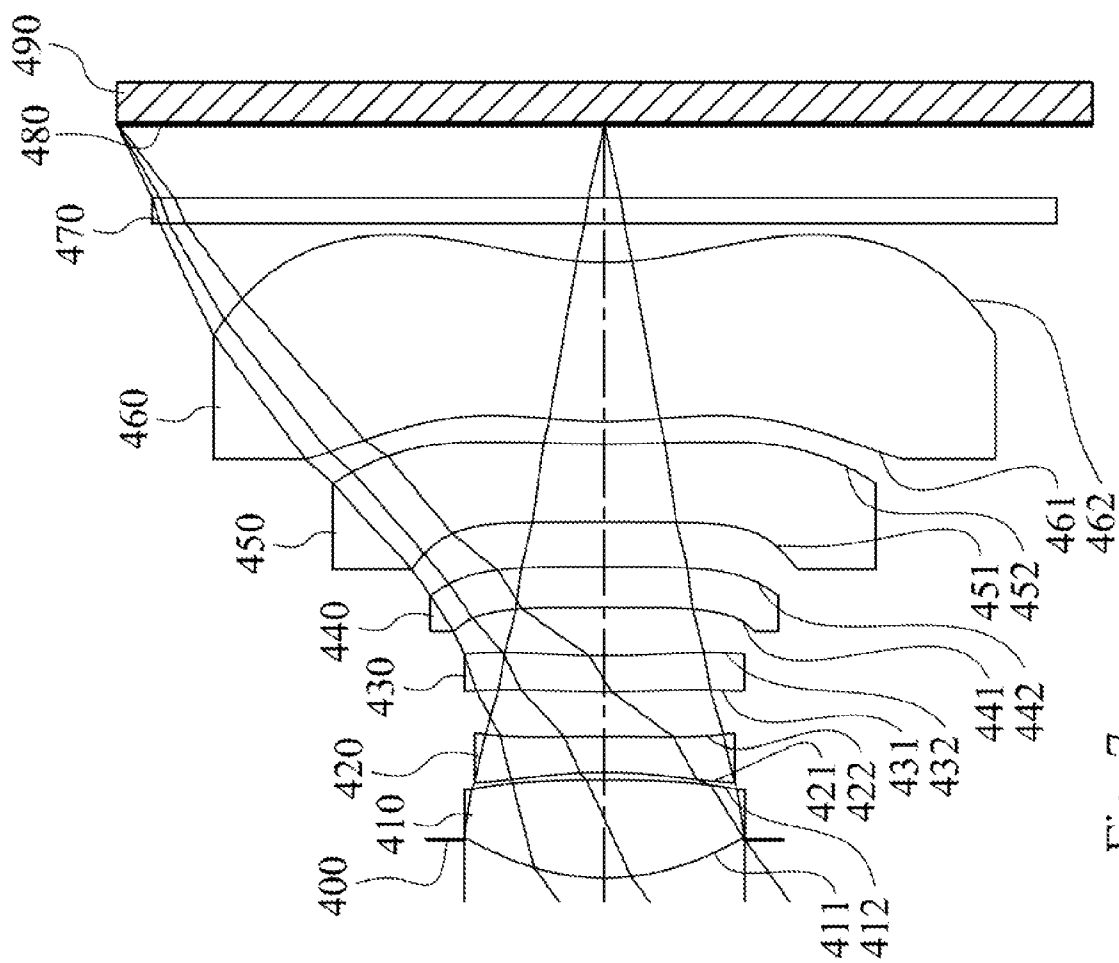


Fig. 7

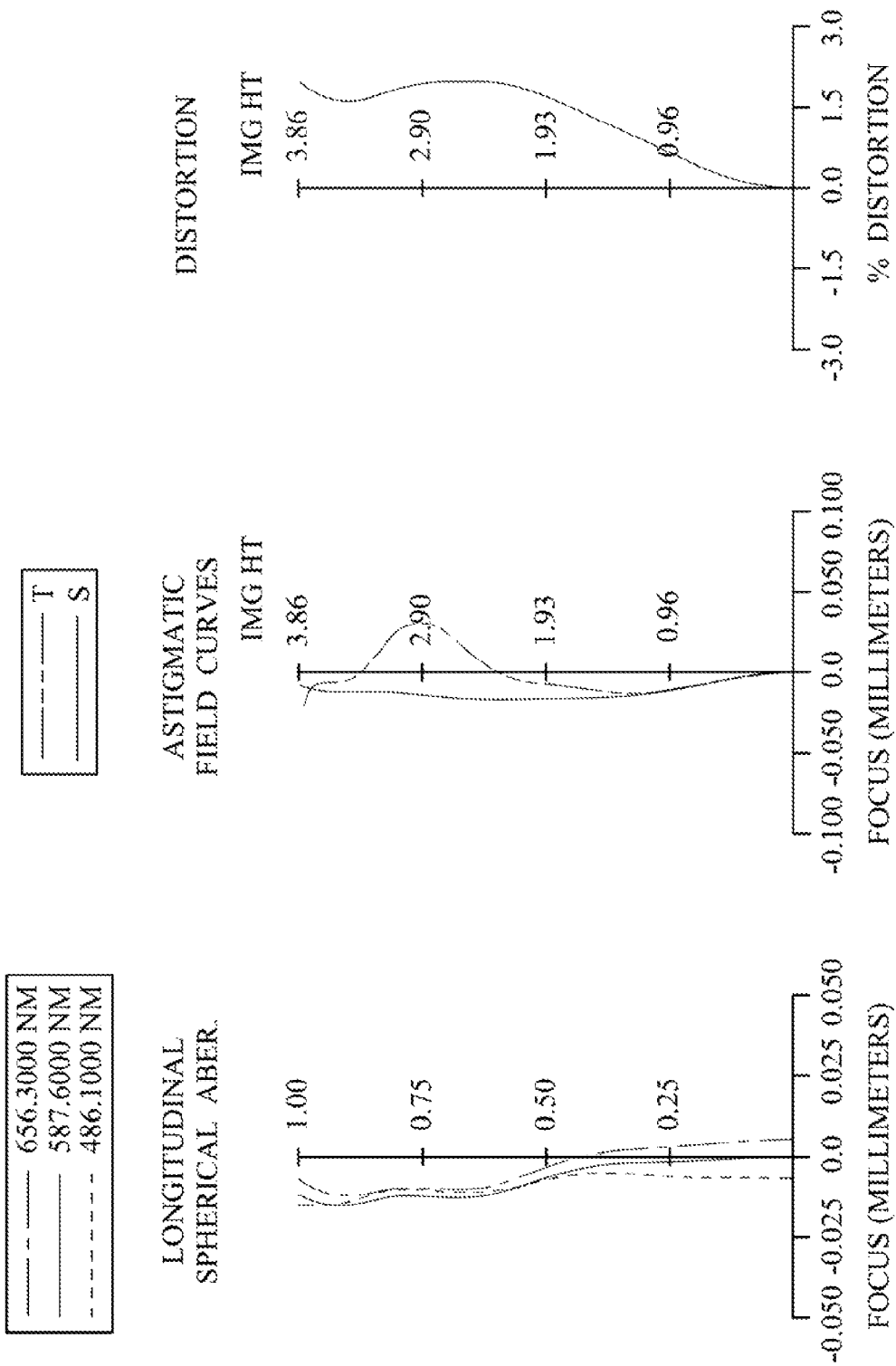


Fig. 8

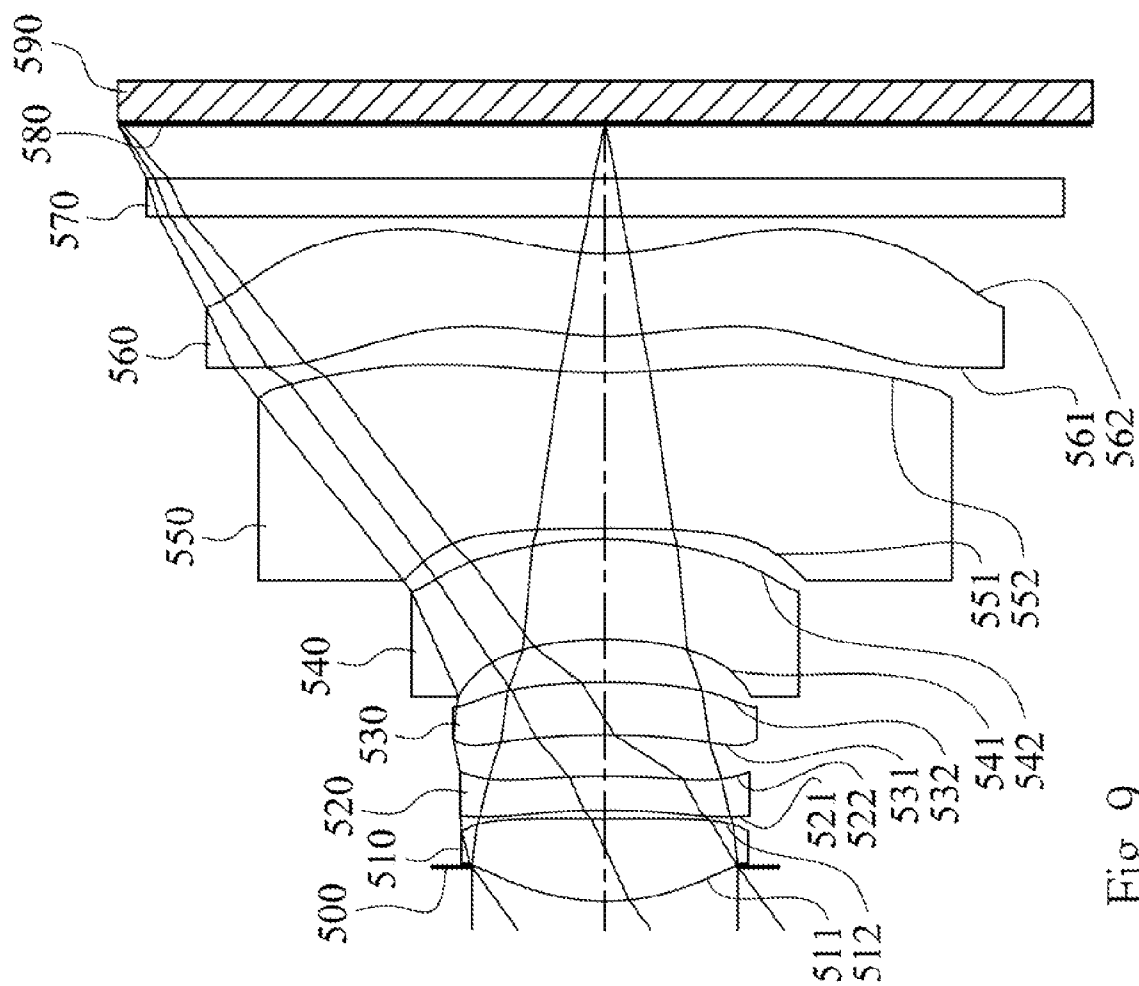


Fig. 9

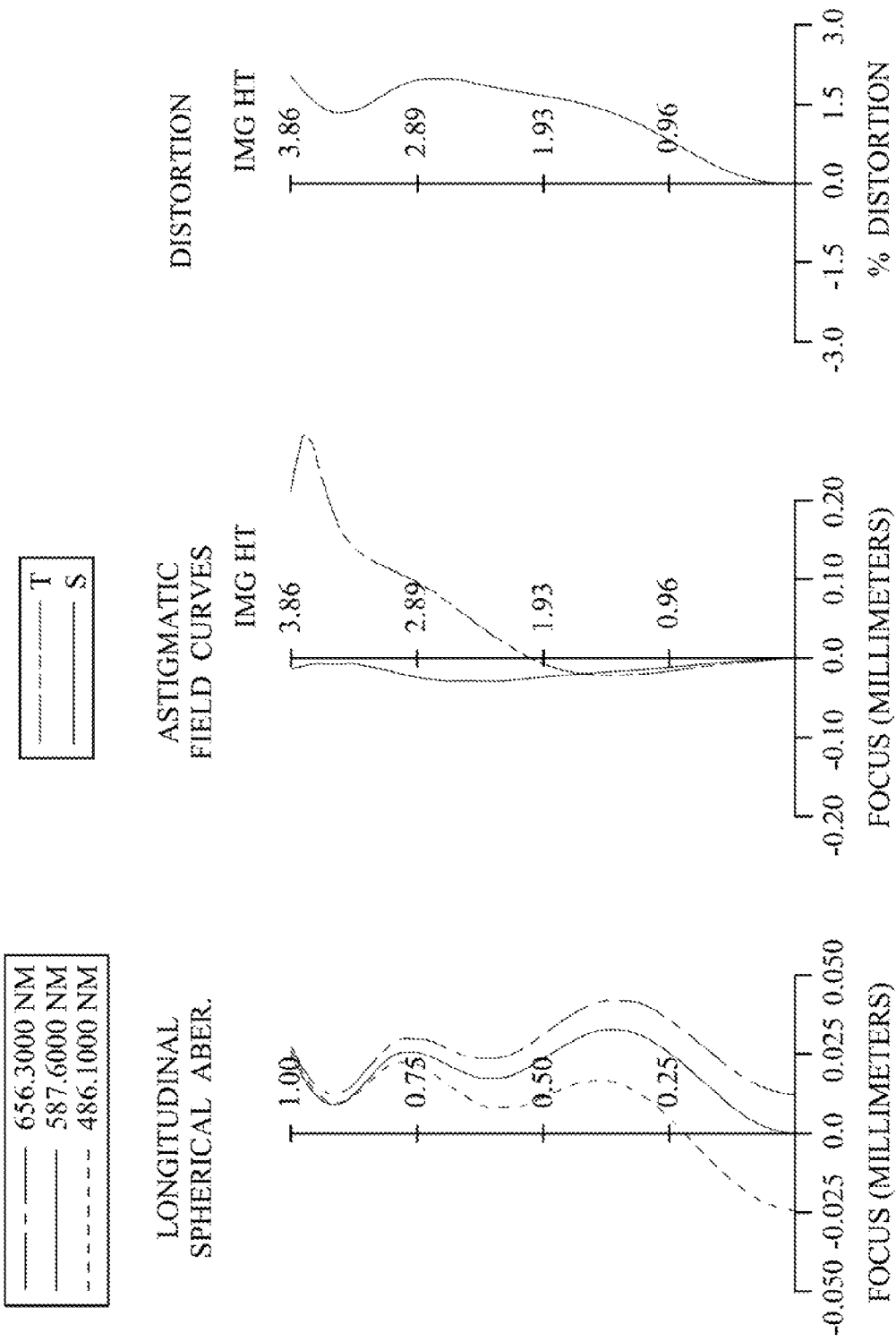


Fig. 10

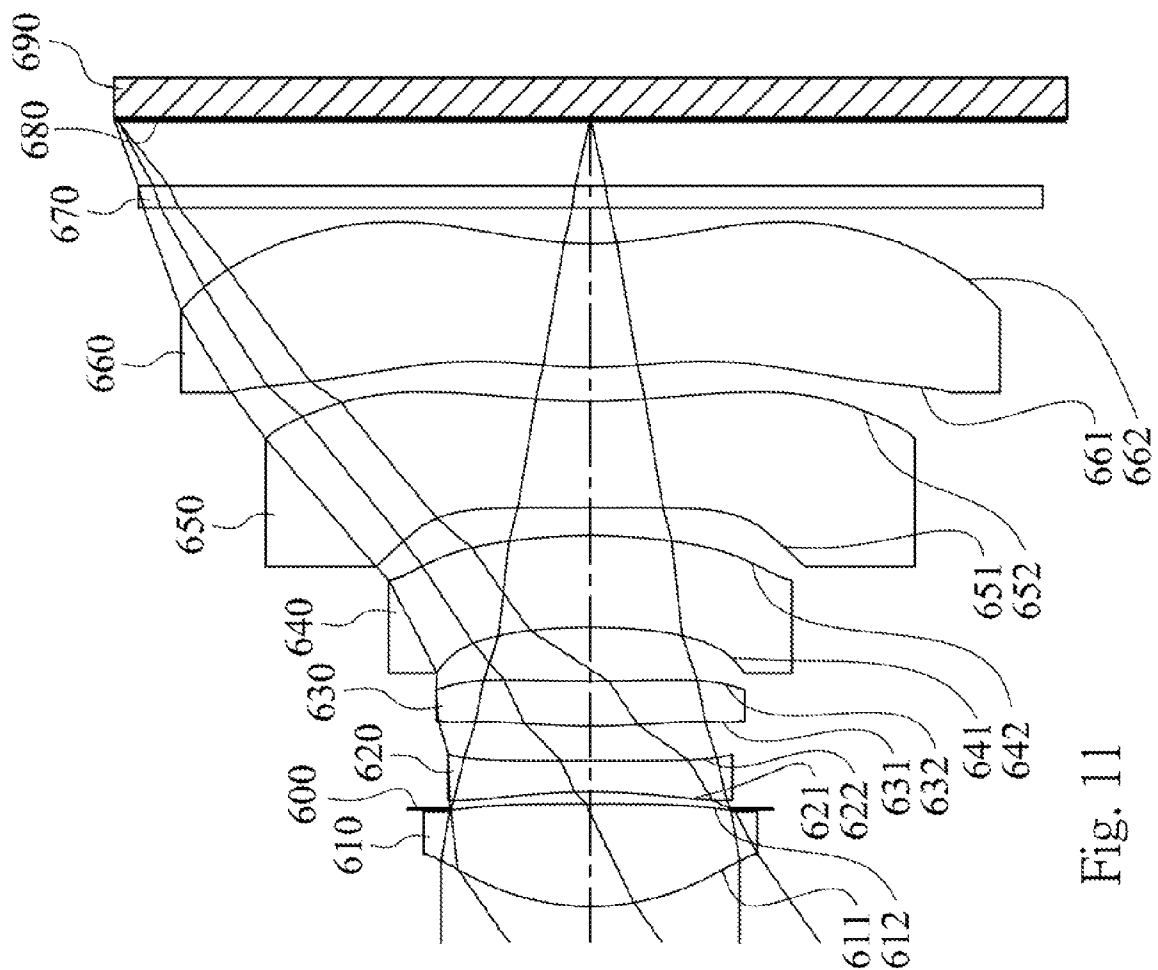


Fig. 11

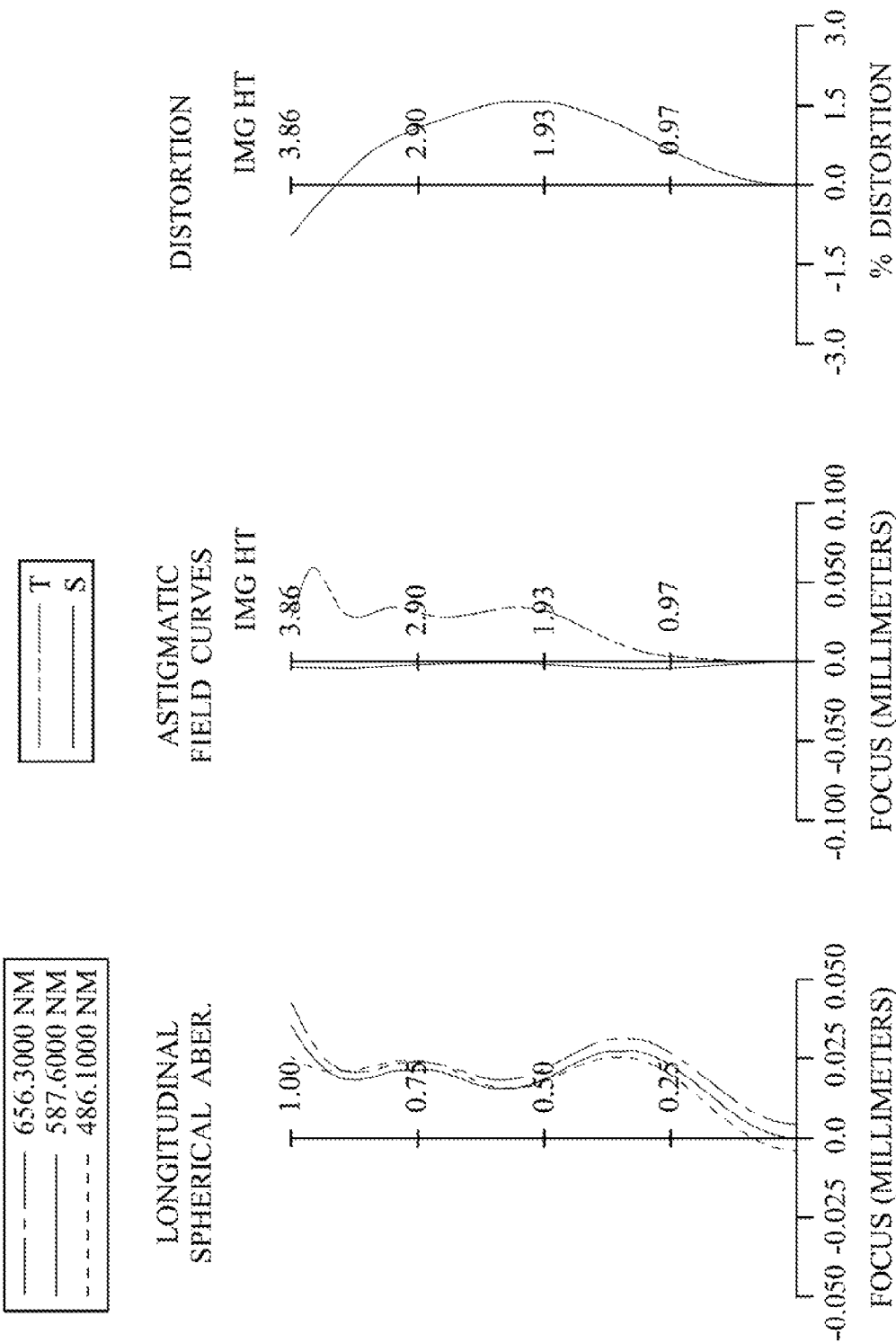


Fig. 12

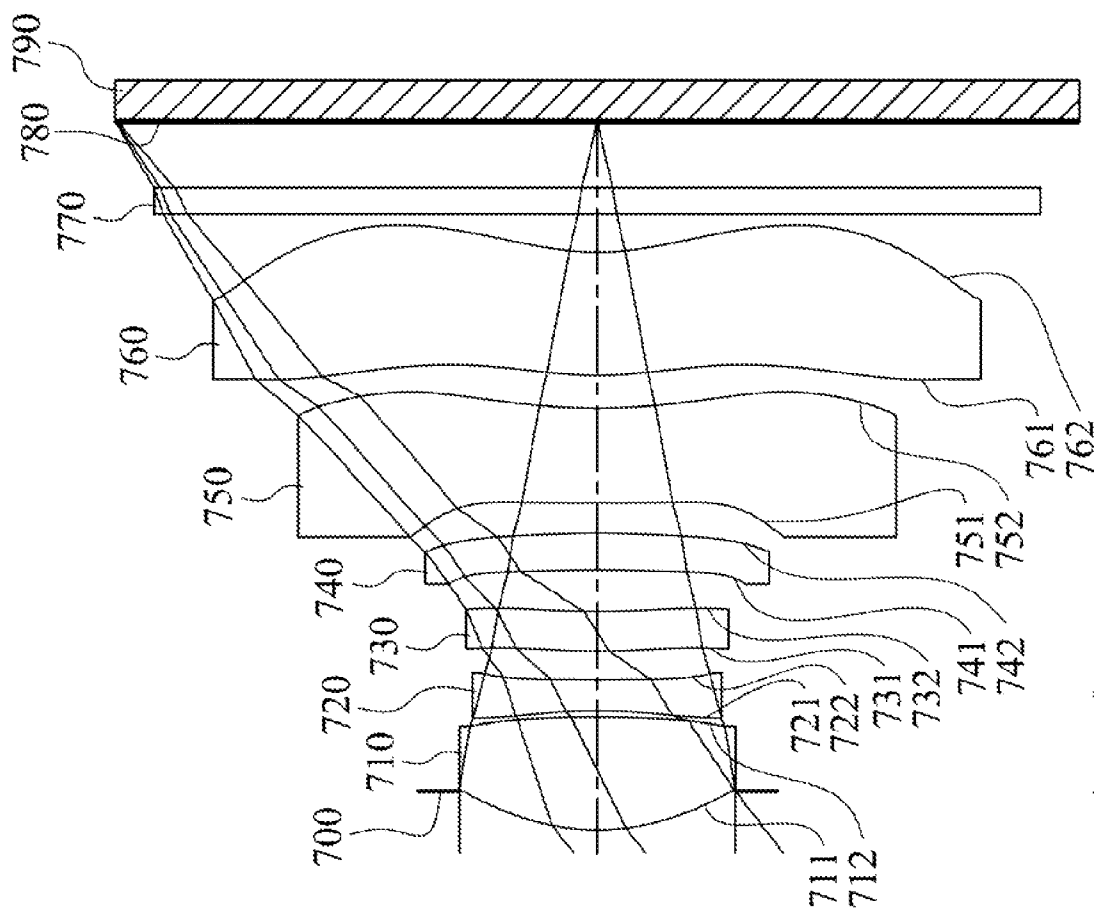


Fig. 13

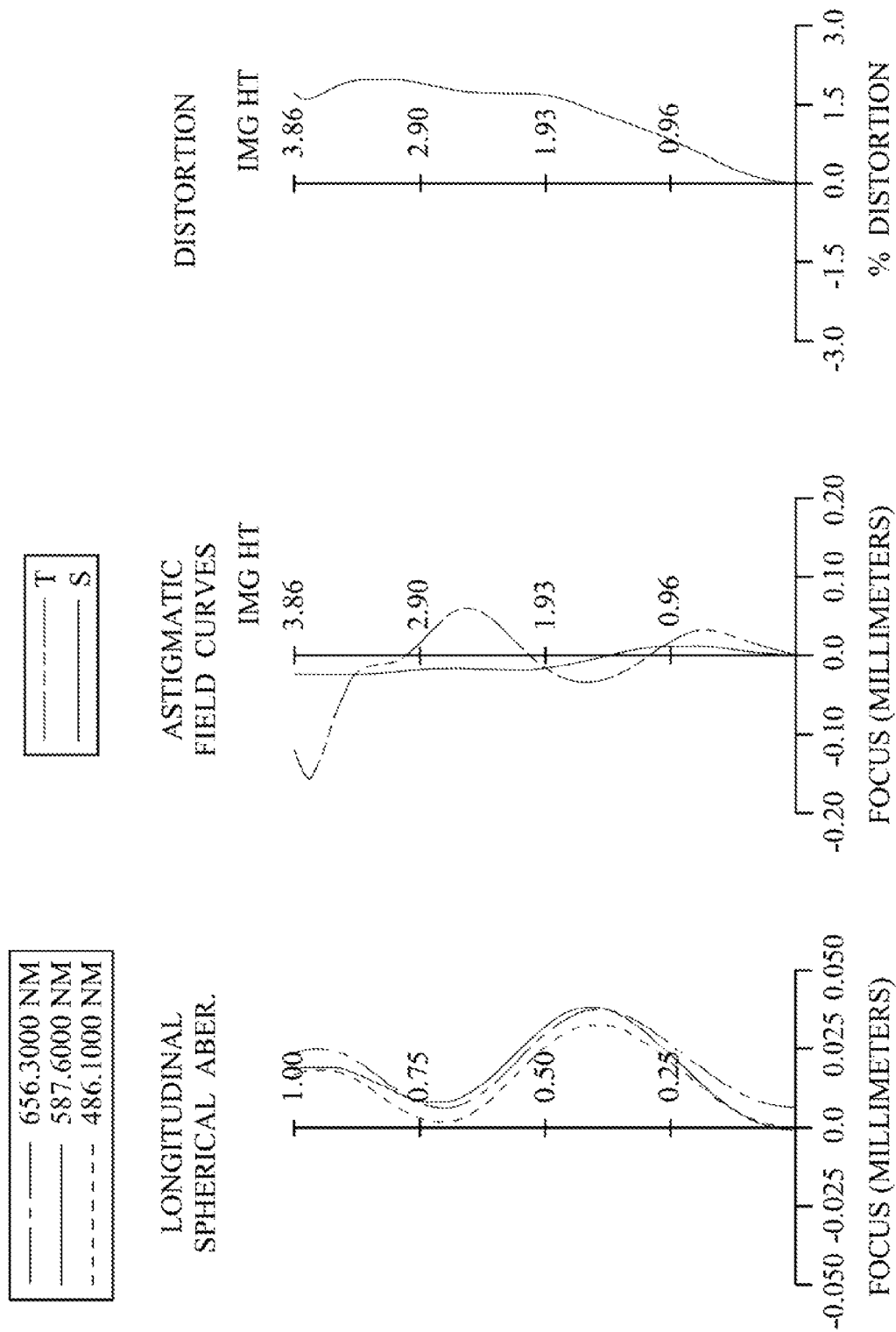


Fig. 14

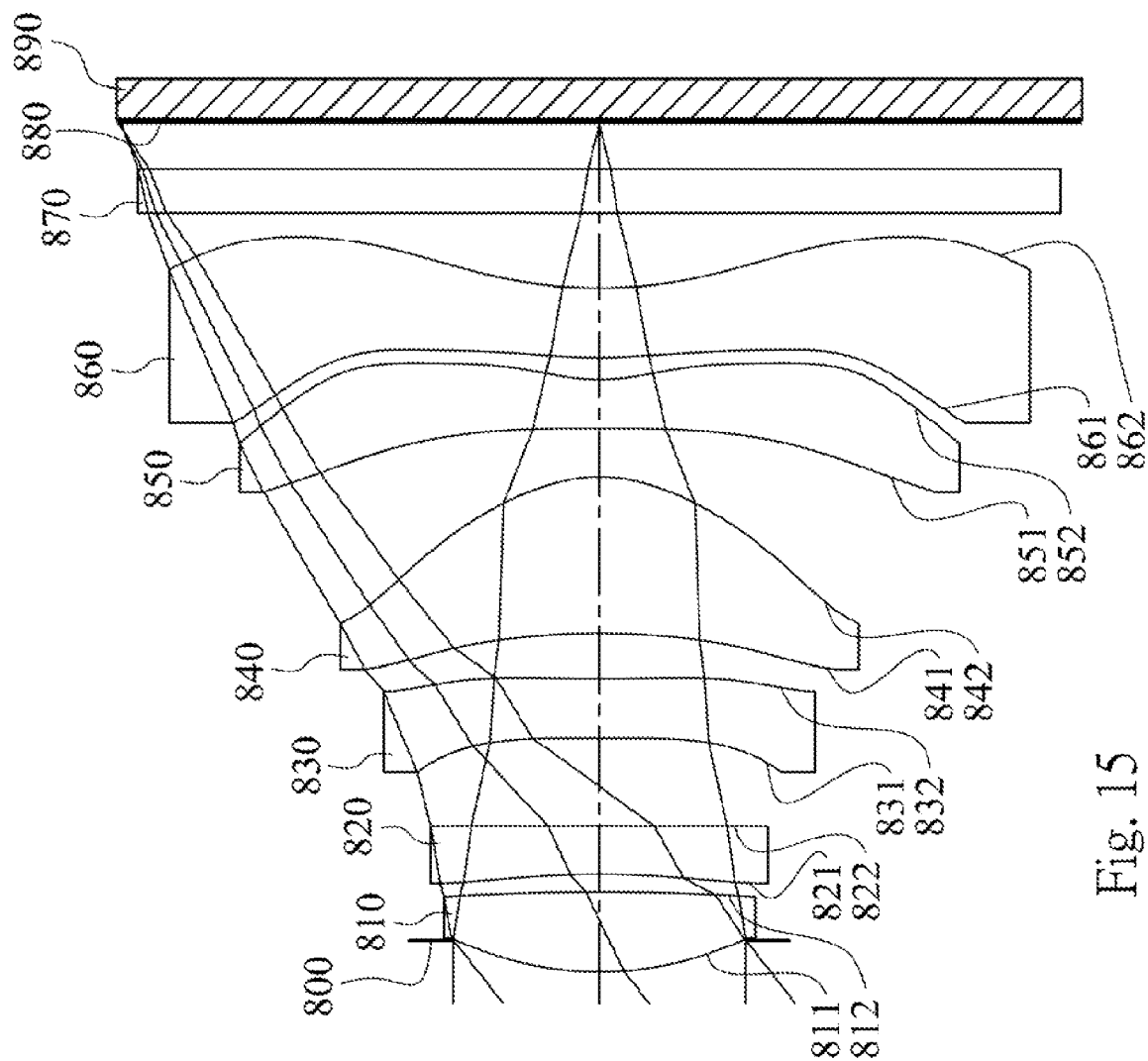
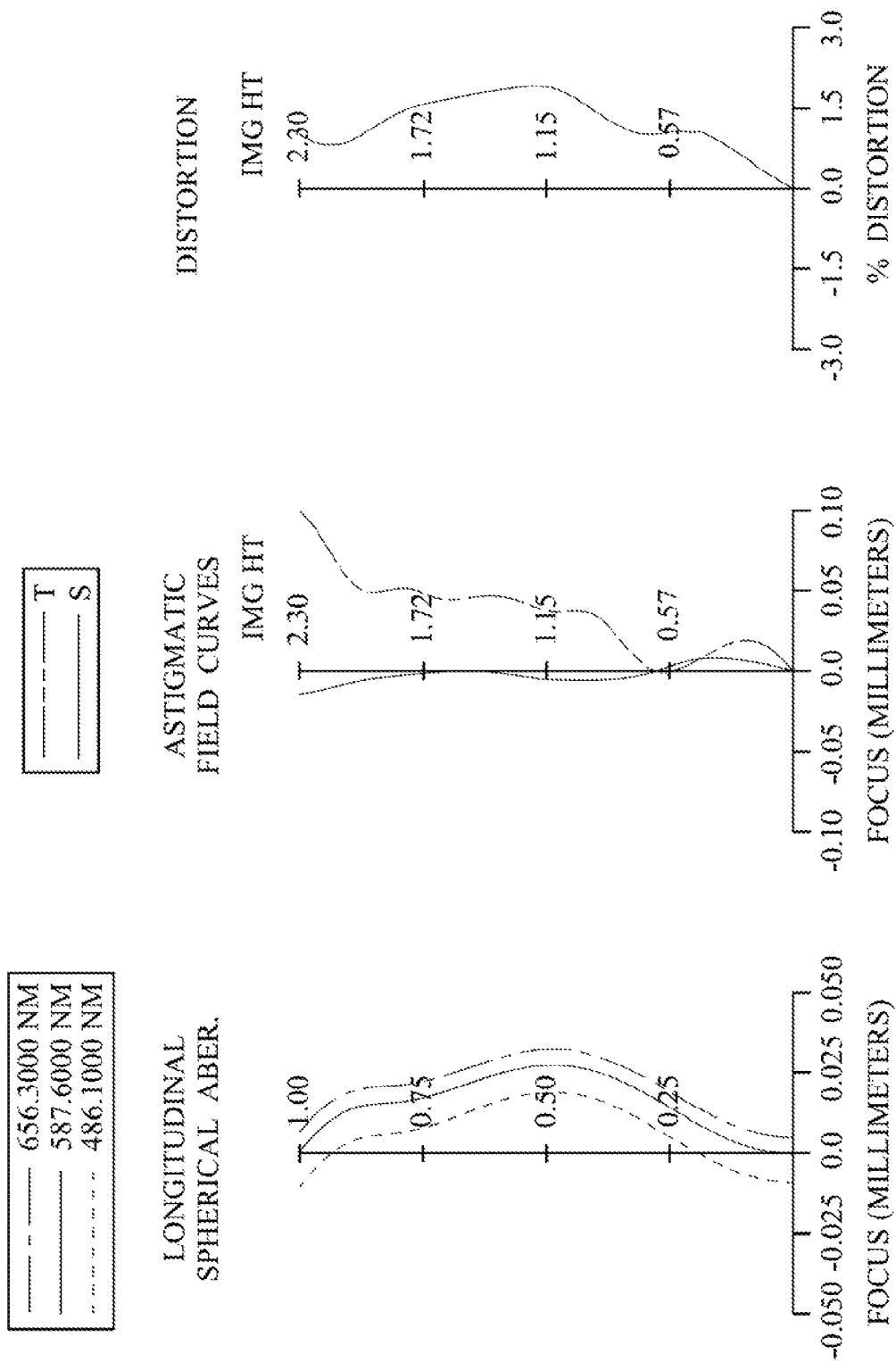


Fig. 15



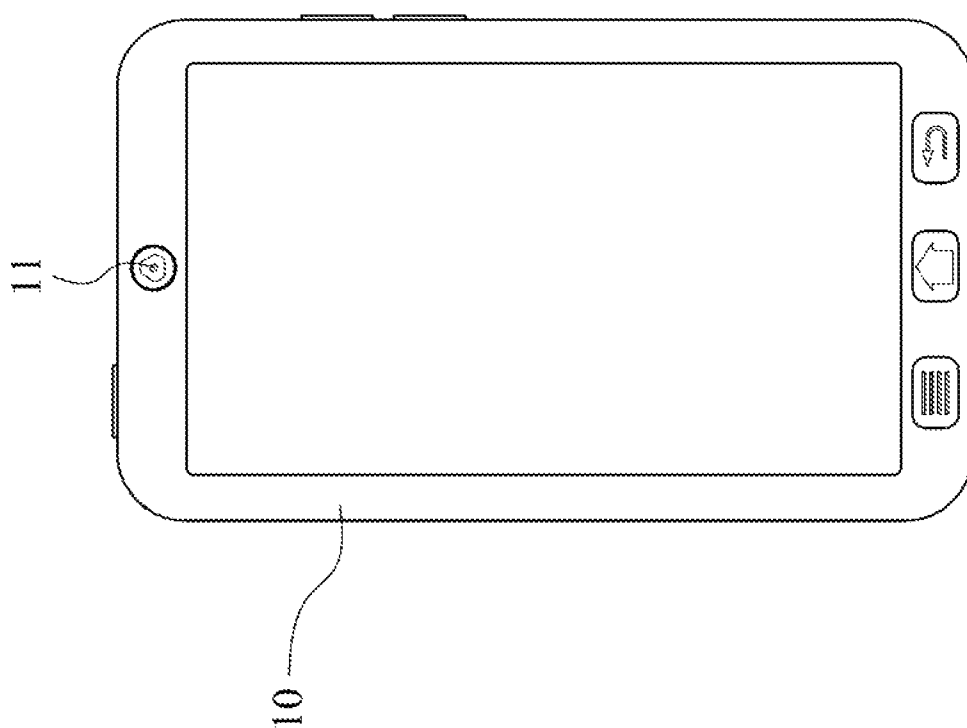


Fig. 17

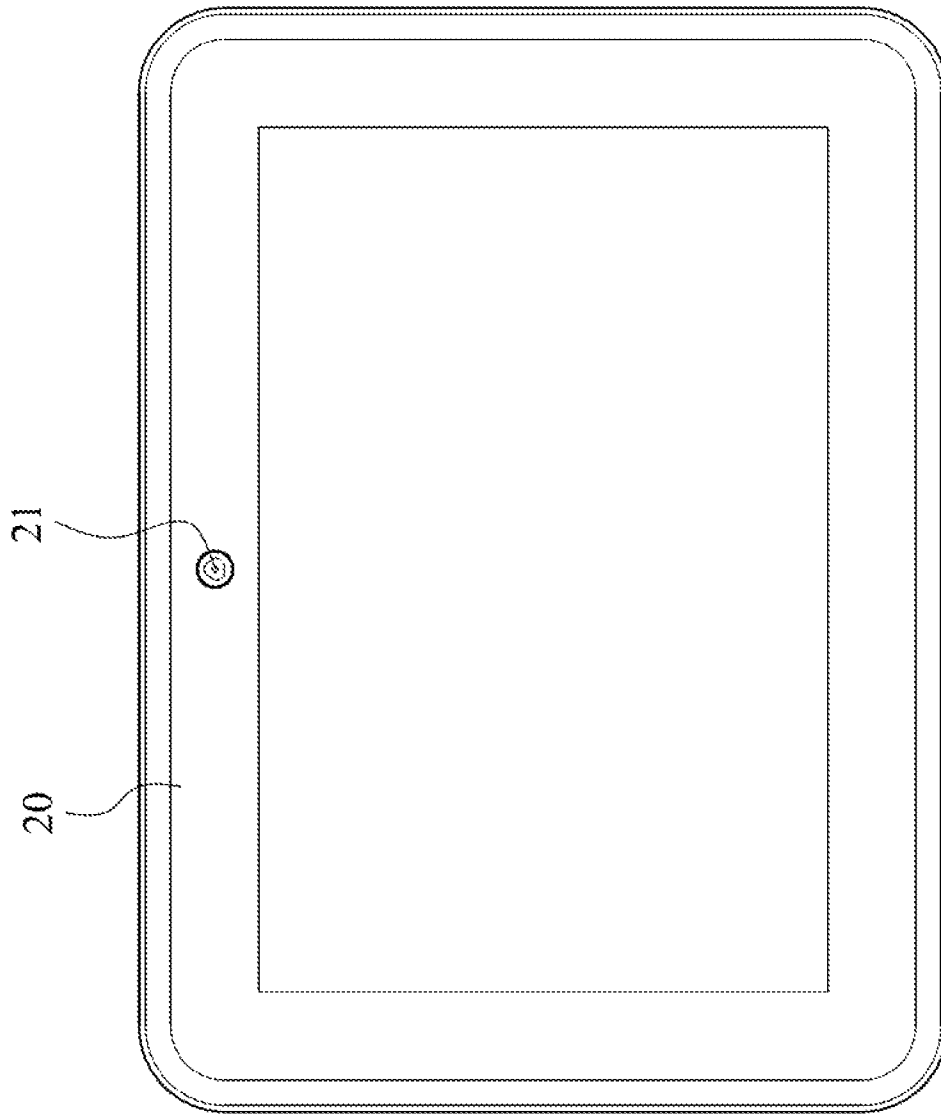


Fig. 18

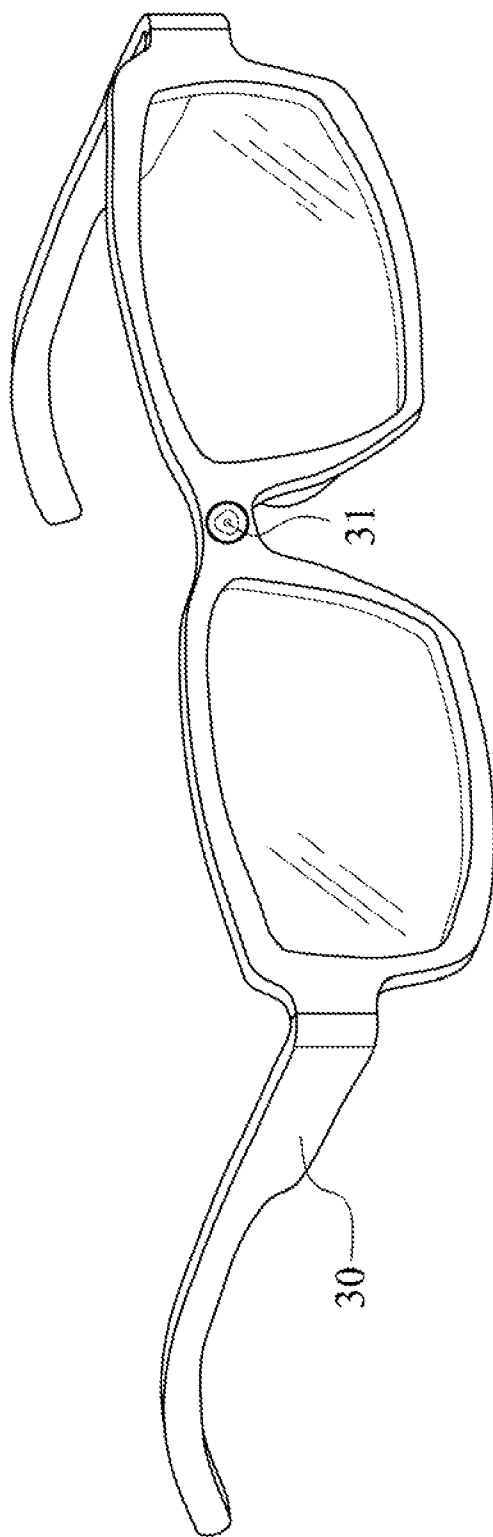


Fig. 19

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PHOTOGRAPHING LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND MOBILE TERMINAL

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 102142878, filed Nov. 25, 2013, which is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to a photographing lens assembly. More particularly, the present disclosure relates to a compact photographing lens assembly applicable to a mobile terminal.

2. Description of Related Art

In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

A conventional optical system employed in a portable electronic product mainly adopts a four-element lens structure or a five-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. However, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

Other conventional compact optical systems with six-element lens structure enhance image quality and resolution. However, the surface shape and refractive power of the lens elements are not favorable for reducing the back focal length of the photographing lens assembly and thus not favorable for keeping a compact size thereof.

SUMMARY

According to one aspect of the present disclosure, a photographing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element with refractive power has an object-side surface being concave in a paraxial region thereof. The third lens element has refractive power. The fourth lens element has refractive power. The fifth lens element with negative refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The photographing lens assembly has a total of six non-cemented

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lens elements with refractive power. When a focal length of the photographing lens assembly is f , a focal length of the first lens element is f_1 , a curvature radius of the object-side surface of the second lens element is R_3 , and a curvature radius of an image-side surface of the second lens element is R_4 , the following conditions are satisfied:

$$0 < f_1/f < 2.5; \text{ and}$$

$$(R_3 + R_4)/(R_3 - R_4) < -0.40.$$

According to another aspect of the present disclosure, an image capturing device includes the photographing lens assembly according to the aforementioned aspect and an image sensor, wherein the image sensor is located on an image plane of the photographing lens assembly.

According to still another aspect of the present disclosure, a mobile terminal includes the image capturing device according to the aforementioned aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment;

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

FIG. 17 shows a mobile terminal according to the 9th embodiment;

FIG. 18 shows a mobile terminal according to the 10th embodiment; and

FIG. 19 shows a mobile terminal according to the 11th embodiment.

A photographing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The photographing lens assembly has a total of six non-cemented lens elements with refractive power.

Each of the first through sixth lens elements is a single and non-cemented lens element. That is, any two lens elements adjacent to each other are not cemented, and there is a space in a paraxial region between the two lens elements. Moreover, the manufacturing process of the cemented lenses is more complex than the non-cemented lenses. In particular, a second surface of one lens element and a first surface of the following lens element need to have accurate curvature to ensure these two lens elements will be highly cemented. However, during the cementing process, those two lens elements might not be highly cemented due to displacement and it is thereby not favorable for the image quality of the photographing lens assembly. Therefore, the photographing lens assembly of the present disclosure provides six non-cemented lens elements for improving the problem generated by the cemented lens elements.

The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof, so that the total track length of the photographing lens assembly can be reduced by properly adjusting the positive refractive power of the first lens element.

The second lens element can have negative refractive power and an object-side surface being concave in a paraxial region thereof. Therefore, it is favorable for correcting the aberration of the photographing lens assembly.

The fourth lens element can have positive refractive power, wherein the positive refractive power of the fourth lens element can become weak from a paraxial region towards a peripheral region. Moreover, the fourth lens element can have an object-side surface being concave in a paraxial region thereof and an image-side surface being convex in a paraxial region thereof. Therefore, it is favorable for effectively correcting astigmatism and reducing photosensitivity and further correcting aberration of the off-axis.

The fifth lens element with negative refractive power has an object-side surface being concave in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for the principal point of the photographing lens assembly being positioned away from the image plane and reducing the back focal length thereof so as to keep a compact size. Furthermore, the image-side surface of the fifth lens element can have at least one convex shape in an off-axis region thereof, so that it is favorable for effectively reducing the incident angle of the light projecting onto an image sensor in order to improve the responding efficiency of the image sensor.

The sixth lens element can have an object-side surface being convex in a paraxial region thereof and has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof. Therefore, it is favorable for reducing the back focal length of the photographing lens assembly so as to correct the aberration of the off-axis.

According to the photographing lens assembly of the present disclosure, each of the second lens element, the third lens element and the fourth lens element has at least one inflection point on at least one of the object-side surface and the image-side surface thereof. Therefore, it is favorable for

reducing the incident angle of the light projecting onto the image sensor so as to improve the responding efficiency of the image sensor and reduce the aberration of the off-axis.

When a focal length of the photographing lens assembly is f , and a focal length of the first lens element is f_1 , the following condition is satisfied: $0 < f_1/f < 2.5$. Therefore, it is favorable for reducing the total track length of the photographing lens assembly so as to keep a compact size thereof. Preferably, the following condition is satisfied: $0.30 < f_1/f < 1.25$.

When a curvature radius of the object-side surface of the second lens element is R_3 , and a curvature radius of the image-side surface of the second lens element is R_4 , the following condition is satisfied: $(R_3+R_4)/(R_3-R_4) < -0.40$. Therefore, it is favorable for correcting the aberration of the photographing lens assembly by properly adjusting the surface shape of the second lens element. Preferably, the following condition is satisfied: $-2.0 < (R_3+R_4)/(R_3-R_4) < -0.60$.

When an Abbe number of the fifth lens element is V_5 , the following condition is satisfied: $V_5 < 32$. Therefore, it is favorable for correcting the chromatic aberration of the photographing lens assembly.

When a curvature radius of the object-side surface of the fifth lens element is R_9 , and a curvature radius of the image-side surface of the fifth lens element is R_{10} , the following condition is satisfied: $0 < (R_9+R_{10})/(R_9-R_{10}) < 1.0$. Therefore, it is favorable for having a stronger negative refractive power for the fifth lens element by properly adjusting the surface shape of the fifth lens element. Furthermore, it is also favorable for the principal point of the photographing lens assembly being positioned away from the image plane and reducing the back focal length of the photographing lens assembly so as to keep a compact size thereof.

When the focal length of the photographing lens assembly is f , and the curvature radius of the image-side surface of the second lens element is R_4 , the following condition is satisfied: $-2.0 < f/R_4 \leq 0.0$. Therefore, it is favorable for correcting the aberration of the photographing lens assembly.

When the focal length of the first lens element is f_1 , and a focal length of the second lens element is f_2 , the following condition is satisfied: $f_1/|f_2| < 0.75$. Therefore, it is favorable for balancing the arrangement of the refractive powers.

When an axial distance between the object-side surface of the first lens element and the image plane is TL , and a maximum image height of the photographing lens assembly (half of a diagonal length of an effective photosensitive area of an image sensor) is $ImgH$, the following condition is satisfied: $TL/ImgH < 2.0$. Therefore, it is favorable for keeping the photographing lens assembly compact.

When a curvature radius of the object-side surface of the first lens element is R_1 , and a curvature radius of the image-side surface of the first lens element is R_2 , the following condition is satisfied: $-1.5 < (R_1+R_2)/(R_1-R_2) < -0.4$. Therefore, it is favorable for reducing the spherical aberration and astigmatism of the photographing lens assembly.

When a sum of the central thicknesses of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element is ΣCT , and an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is T_d , the following condition is satisfied: $0.65 < \Sigma CT/T_d < 0.85$. Therefore, it provides favorable moldability and homogeneity for lens elements during the injection molding process and keeping the photographing lens assembly compact.

According to the photographing lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of

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glass material, the distribution of the refractive power of the photographing lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the photographing lens assembly can also be reduced.

According to the photographing lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof.

According to the photographing lens assembly of the present disclosure, the photographing lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the photographing lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the photographing lens assembly and the image plane and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image plane is favorable for enlarging the field of view of the photographing lens assembly and thereby provides a wider field of view for the same.

The present photographing lens assembly can be optionally applied to moving focus optical systems. According to the photographing lens assembly of the present disclosure, the photographing lens assembly is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals.

According to the present disclosure, an image capturing device is provided. The image capturing device includes the photographing lens assembly according to the aforementioned photographing lens assembly of the present disclosure, and an image sensor, wherein the image sensor is disposed on an image plane of the aforementioned photographing lens assembly. In the image capturing device, the lens elements of the photographing lens assembly can have much stronger negative refractive power through the arrangements of the surface shape and refractive powers of the photographing lens assembly. Therefore, it is favorable for the principal point of the photographing lens assembly being positioned away from the image plane so as to reduce the back focal length thereof and be applied to compact mobile terminals. Preferably, the image capturing device can further include a barrel member, a holding member or a combination thereof.

According to the present disclosure, a mobile terminal is provided, wherein the mobile terminal includes the aforementioned image capturing device. The image capturing device includes the photographing lens assembly according

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to the aforementioned photographing lens assembly of the present disclosure, and the image sensor, wherein the image sensor is disposed on an image plane of the aforementioned photographing lens assembly. Therefore, it is favorable for maintaining a compact size thereof. Preferably, the mobile terminal can further include but not limited to display, control unit, random access memory unit (RAM) a read only memory unit (ROM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-11th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment.

In FIG. 1, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 190. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an image plane 180, wherein the photographing lens assembly has a total of six non-cemented lens elements (110-160) with refractive power.

The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being convex in a paraxial region thereof. The first lens element 110 is made of plastic material and has the object-side surface 111 and the image-side surface 112 being both aspheric.

The second lens element 120 with negative refractive power has an object-side surface 121 being concave in a paraxial region thereof and an image-side surface 122 being convex in a paraxial region thereof. The second lens element 120 is made of plastic material and has the object-side surface 121 and the image-side surface 122 being both aspheric, wherein both of the object-side surface 121 and the image-side surface 122 of the second lens element 120 have at least one inflection point.

The third lens element 130 with negative refractive power has an object-side surface 131 being concave in a paraxial region thereof and an image-side surface 132 being concave in a paraxial region thereof. The third lens element 130 is made of plastic material and has the object-side surface 131 and the image-side surface 132 being both aspheric, wherein the image-side surface 132 of the third lens element 130 has at least one inflection point.

The fourth lens element 140 with positive refractive power has an object-side surface 141 being concave in a paraxial region thereof and an image-side surface 142 being convex in a paraxial region thereof. The fourth lens element 140 is made of plastic material and has the object-side surface 141 and the image-side surface 142 being both aspheric, wherein both of the object-side surface 141 and the image-side surface 142 of the fourth lens element 140 have at least one inflection point. Furthermore, the positive refractive power of the fourth lens element 140 becomes weak from a paraxial region towards a peripheral region.

The fifth lens element 150 with negative refractive power has an object-side surface 151 being concave in a paraxial region thereof and an image-side surface 152 being concave in a paraxial region thereof. The fifth lens element 150 is

made of plastic material and has the object-side surface **151** and the image-side surface **152** being both aspheric, wherein the image-side surface **152** of the fifth lens element **150** has at least one convex shape in an off-axis region thereof.

The sixth lens element **160** with positive refractive power has an object-side surface **161** being convex in a paraxial region thereof and an image-side surface **162** being concave in a paraxial region thereof, wherein the image-side surface **162** of the sixth lens element **160** has at least one convex shape in an off-axis region thereof. The sixth lens element **160** is made of plastic material and has the object-side surface **161** and the image-side surface **162** being both aspheric.

The IR-cut filter **170** is made of glass and located between the sixth lens element **160** and the image plane **180**, and will not affect the focal length of the photographing lens assembly. The image sensor **190** is disposed on the image plane **180** of the photographing lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y')^i,$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when a focal length of the photographing lens assembly is f, an f-number of the photographing lens assembly is Fno, and half of a maximal field of view of the photographing lens assembly is HFOV, these parameters have the following values: f=3.13 mm; Fno=2.20; and HFOV=36.0 degrees.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when an Abbe number of the fifth lens element **150** is V5, and the following condition is satisfied: V5=55.9.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when a curvature radius of the object-side surface **111** of the first lens element **110** is R1, a curvature radius of the image-side surface **112** of the first lens element **110** is R2, a curvature radius of the object-side surface **121** of the second lens element **120** is R3, a curvature radius of the image-side surface **122** of the second lens element **120** is R4, a curvature radius of the object-side surface **151** of the fifth lens element **150** is R9, and a curvature radius of the image-side surface **152** of the fifth lens element **150** is R10, the following conditions are satisfied: (R1+R2)/(R1-R2)=-0.55; (R3+R4)/(R3-R4)=-1.26; and (R9+R10)/(R9-R10)=0.89.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when the focal length of the photographing lens assembly is f, and the curvature radius of the image-side surface **122** of the second lens element **120** is R4, the following condition is satisfied: f/R4=-0.13.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when the focal length of the photographing lens assembly is f, a focal length of the first lens element **110** is f1, and a focal length of the second lens element **120** is f2, the following conditions are satisfied: f1/f=0.72; and f1/|f2|=0.43.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when a sum of the central thicknesses of the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150**, and the sixth lens element **160** is ΣCT, and an axial distance between the object-side surface **111** of the first lens element **110** and the image-side surface **162** of the sixth lens element **160** is Td, the following condition is satisfied: ΣCT/Td=0.67.

In the photographing lens assembly of the image capturing device according to the 1st embodiment, when an axial distance between the object-side surface **111** of the first lens element **110** and the image plane **180** is TL, and a maximum image height of the photographing lens assembly (half of a diagonal length of an effective photosensitive area of the image sensor **190**) is ImgH, the following condition is satisfied: TL/ImgH=1.81.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment								
f = 3.13 mm, Fno = 2.20, HFOV = 36.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.160					
2	Lens 1	1.543 ASP	0.463	Plastic	1.544	55.9	2.25	
3		-5.295 ASP	0.065					
4	Lens 2	-2.914 ASP	0.230	Plastic	1.634	23.8	-5.22	
5		-25.000 ASP	0.516					
6	Lens 3	-52.617 ASP	0.300	Plastic	1.634	23.8	-15.69	
7		12.297 ASP	0.175					
8	Lens 4	-3.350 ASP	0.755	Plastic	1.544	55.9	1.69	
9		-0.778 ASP	0.256					
10	Lens 5	-16.129 ASP	0.237	Plastic	1.544	55.9	-1.60	
11		0.924 ASP	0.104					
12	Lens 6	2.427 ASP	0.250	Plastic	1.614	25.6	126.42	
13		2.407 ASP	0.360					

TABLE 1-continued

1st Embodiment						
f = 3.13 mm, Fno = 2.20, HFOV = 36.0 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Focal Length
14	IR- cut filter	Plano	0.210	Glass	1.517	64.2
15		Plano	0.230			
16	Image	Plano	—			

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-5.5978E+00	-4.2085E+01	-5.2713E+01	-1.0000E+00	9.0000E+01	9.0000E+01
A4 =	1.6340E-01	4.0145E-02	-5.6465E-02	1.3074E-01	-1.6335E-01	-1.0387E-01
A6 =	-2.4866E-02	1.0504E-01	5.8559E-01	-4.1045E-01	-8.2226E-01	-2.9160E-01
A8 =	-3.2822E-01	-8.4012E-01	-2.0777E+00	1.1606E+00	3.2179E+00	7.5402E-01
A10 =	8.1580E-01	2.2212E+00	4.2635E+00	-2.3888E+00	-7.5527E+00	-1.1540E+00
A12 =	-1.0388E+00	-3.0026E+00	-4.9172E+00	2.8633E+00	1.0182E+01	1.1528E+00
A14 =	6.4166E-01	1.8910E+00	2.8800E+00	-1.8786E+00	-7.0990E+00	-5.7390E-01
A16 =	-1.9869E-01	-4.6782E-01	-6.4826E-01	5.1494E-01	1.9702E+00	1.0878E-01
Surface #	8	9	10	11	12	13
k =	4.6824E+00	-3.9998E+00	4.5273E+01	-6.5922E+00	-8.7439E-02	-2.0250E-01
A4 =	2.2425E-02	-3.0034E-01	-3.3194E-01	-6.0022E-01	-2.2864E-01	7.8256E-02
A6 =	1.1070E-01	5.6210E-01	6.3301E-01	9.3430E-01	1.8505E-01	-2.3435E-01
A8 =	-4.2143E-01	-8.4462E-01	-8.1791E-01	-7.7884E-01	-1.0148E-01	1.7002E-01
A10 =	5.1296E-01	7.6007E-01	6.0538E-01	3.6064E-01	2.4404E-02	-6.5297E-02
A12 =	-2.2753E-01	-3.8589E-01	-2.4630E-01	-9.3891E-02	-1.5152E-03	1.3742E-02
A14 =	1.9640E-02	1.0771E-01	5.1356E-02	1.2771E-02	-2.5015E-04	-1.4808E-03
A16 =	7.6264E-03	-1.3199E-02	-4.2880E-03	-7.0249E-04	2.8553E-05	6.3462E-05

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment.

In FIG. 3, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 290. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 200, a first lens element 210, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, a sixth lens element 260, an IR-cut filter 270 and an image plane 280, wherein the photographing lens assembly has a total of six non-cemented lens elements (210-260) with refractive power.

The first lens element 210 with positive refractive power has an object-side surface 211 being convex in a paraxial

region thereof and an image-side surface 212 being convex in a paraxial region thereof. The first lens element 210 is made of plastic material and has the object-side surface 211 and the image-side surface 212 being both aspheric.

The second lens element 220 with negative refractive power has an object-side surface 221 being concave in a paraxial region thereof and an image-side surface 222 being convex in a paraxial region thereof. The second lens element 220 is made of plastic material and has the object-side surface 221 and the image-side surface 222 being both aspheric.

The third lens element 230 with positive refractive power has an object-side surface 231 being convex in a paraxial region thereof and an image-side surface 232 being concave in a paraxial region thereof. The third lens element 230 is made of plastic material and has the object-side surface 231 and the image-side surface 232 being both aspheric, wherein both of the object-side surface 231 and the image-side surface 232 of the third lens element 230 have at least one inflection point.

The fourth lens element 240 with positive refractive power has an object-side surface 241 being concave in a paraxial region thereof and an image-side surface 242 being convex in a paraxial region thereof. The fourth lens element 240 is made of plastic material and has the object-side surface 241 and the image-side surface 242 being both aspheric, wherein both of the object-side surface 241 and the image-side surface 242 of the fourth lens element 240 have at least one inflection point. Furthermore, the positive refractive power of the fourth lens element 240 becomes weak from a paraxial region towards a peripheral region.

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The fifth lens element **250** with negative refractive power has an object-side surface **251** being concave in a paraxial region thereof and an image-side surface **252** being concave in a paraxial region thereof. The fifth lens element **250** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric, wherein the image-side surface **252** of the fifth lens element **250** has at least one convex shape in an off-axis region thereof.

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The IR-cut filter **270** is made of glass and located between the sixth lens element **260** and the image plane **280**, and will not affect the focal length of the photographing lens assembly. The image sensor **290** is disposed on the image plane **280** of the photographing lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 3.14 mm, Fno = 2.20, HFOV = 36.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.132					
2	Lens 1	1.468 ASP	0.592	Plastic	1.544	55.9	2.23	
3		-6.064 ASP	0.059					
4	Lens 2	-2.501 ASP	0.230	Plastic	1.634	23.8	-5.01	
5		-12.233 ASP	0.377					
6	Lens 3	9.276 ASP	0.252	Plastic	1.634	23.8	55.61	
7		12.456 ASP	0.231					
8	Lens 4	-2.153 ASP	0.863	Plastic	1.544	55.9	1.52	
9		-0.682 ASP	0.161					
10	Lens 5	-8.598 ASP	0.244	Plastic	1.555	36.3	-1.53	
11		0.951 ASP	0.091					
12	Lens 6	2.583 ASP	0.250	Plastic	1.614	25.6	-13.11	
13		1.884 ASP	0.360					
14	IR- cut filter	Plano	0.210	Glass	1.517	64.2	—	
15		Plano	0.234					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-4.3989E+00	-9.0000E+01	-4.2438E+01	-1.0000E+00	-5.9841E+01	5.3960E+01
A4 =	1.5669E-01	1.8082E-02	-7.6133E-02	1.3488E-01	-1.7552E-01	-9.2805E-02
A6 =	-3.0009E-02	5.1895E-02	5.7183E-01	-4.3760E-01	-8.8493E-01	-3.0122E-01
A8 =	-3.0640E-01	-8.6886E-01	-2.1055E+00	1.1243E+00	3.0501E+00	7.6383E-01
A10 =	8.1572E-01	2.2042E+00	4.2257E+00	-2.3828E+00	-7.5988E+00	-1.1459E+00
A12 =	-1.0658E+00	-3.0319E+00	-4.9317E+00	2.8910E+00	1.0247E+01	1.1573E+00
A14 =	6.2370E-01	1.8568E+00	2.9184E+00	-1.8882E+00	-7.0617E+00	-5.7211E-01
A16 =	-2.4369E-01	-3.4851E-01	-6.2459E-01	4.2267E-01	1.8139E+00	1.0572E-01
Surface #	8	9	10	11	12	13
k =	-7.9894E-01	-4.2519E+00	-5.9525E+01	-1.1081E+01	7.7366E-03	-3.6443E-01
A4 =	8.1209E-03	-4.1283E-01	-2.9036E-01	-3.6603E-01	-1.9987E-01	-1.0062E-01
A6 =	1.5484E-02	7.8749E-01	4.6740E-01	5.3583E-01	1.7819E-01	-2.4222E-02
A8 =	5.9591E-03	-1.2411E+00	-5.6216E-01	-3.9322E-01	-7.7028E-02	4.9863E-02
A10 =	2.6931E-03	1.2479E+00	3.9392E-01	1.5937E-01	8.4384E-03	-2.8291E-02
A12 =	3.7597E-04	-7.6647E-01	-1.5130E-01	-3.6639E-02	2.3977E-03	7.2607E-03
A14 =	6.9673E-04	2.6252E-01	2.9600E-02	4.4070E-03	-6.5069E-04	-8.5348E-04
A16 =	9.4270E-04	-3.7591E-02	-2.3067E-03	-2.1298E-04	4.2224E-05	3.6250E-05

The sixth lens element **260** with negative refractive power has an object-side surface **261** being convex in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof, wherein the image-side surface **262** of the sixth lens element **260** has at least one convex shape in an off-axis region thereof. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being both aspheric.

In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

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Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	3.14	$(R9 + R10)/(R9 - R10)$	0.80
Fno	2.20	f/R4	-0.26
HFOV [deg.]	36.0	f1/f	0.71
V5	36.3	f1/f2	0.45
$(R1 + R2)/(R1 - R2)$	-0.61	$\Sigma CT/Td$	0.73
$(R3 + R4)/(R3 - R4)$	-1.51	TL/ImgH	1.81

3rd Embodiment

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment.

In FIG. 5, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 390. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, an IR-cut filter 370 and an image plane 380, wherein the photographing lens assembly has a total of six non-cemented lens elements (310-360) with refractive power.

The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being planar in a paraxial region thereof. The first lens element 310 is made of glass material and has the object-side surface 311 and the image-side surface 312 being both aspheric.

The second lens element 320 with negative refractive power has an object-side surface 321 being concave in a paraxial region thereof and an image-side surface 322 being convex in a paraxial region thereof. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being both aspheric, wherein both of the object-side surface 321 and the image-side surface 322 of the second lens element 320 have at least one inflection point.

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The third lens element 330 with positive refractive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being concave in a paraxial region thereof. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being both aspheric, wherein both of the object-side surface 331 and the image-side surface 332 of the third lens element 330 have at least one inflection point.

The fourth lens element 340 with positive refractive power has an object-side surface 341 being concave in a paraxial region thereof and an image-side surface 342 being convex in a paraxial region thereof. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being both aspheric, wherein the image-side surface 342 of the fourth lens element 340 has at least one inflection point. Furthermore, the positive refractive power of the fourth lens element 340 becomes weak from a paraxial region towards a peripheral region.

The fifth lens element 350 with negative refractive power has an object-side surface 351 being concave in a paraxial region thereof and an image-side surface 352 being concave in a paraxial region thereof. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being both aspheric, wherein the image-side surface 352 of the fifth lens element 350 has at least one convex shape in an off-axis region thereof.

The sixth lens element 360 with negative refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof, wherein the image-side surface 362 of the sixth lens element 360 has at least one convex shape in an off-axis region thereof. The sixth lens element 360 is made of plastic material and has the object-side surface 361 and the image-side surface 362 being both aspheric.

The IR-cut filter 370 is made of glass and located between the sixth lens element 360 and the image plane 380, and will not affect the focal length of the photographing lens assembly. The image sensor 390 is disposed on the image plane 380 of the photographing lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment								
f = 4.83 mm, Fno = 2.20, HFOV = 38.1 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.339				
2	Lens 1	1.785	ASP	0.791	Glass	1.542	62.9	3.29
3		∞	ASP	0.051				
4	Lens 2	-4.817	ASP	0.250	Plastic	1.614	25.6	-8.30
5		-89.145	ASP	0.293				
6	Lens 3	5.908	ASP	0.342	Plastic	1.544	55.9	73.58
7		6.790	ASP	0.380				
8	Lens 4	-8.449	ASP	0.340	Plastic	1.634	23.8	17.72
9		-4.898	ASP	0.228				
10	Lens 5	-19.291	ASP	0.741	Plastic	1.583	30.2	-6.54
11		4.823	ASP	0.193				
12	Lens 6	2.590	ASP	0.963	Plastic	1.535	55.7	-102.02
13		2.152	ASP	0.300				

TABLE 5-continued

3rd Embodiment						
f = 4.83 mm, Fno = 2.20, HFOV = 38.1 deg.						
Surface #		Curvature Radius	Thickness	Material	Index Abbe #	Focal Length
14	IR- cut filter	Plano	0.210	Glass	1.517 64.2	—
15		Plano	0.531			
16	Image	Plano	—			

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 10 is 1.570 mm.

TABLE 6

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	3.2213E-02	-9.0000E+01	-8.7442E+01	-1.0000E+00	-6.4905E+01	3.8823E-01
A4 =	-3.7484E-03	-8.2780E-02	-1.2063E-01	2.9102E-02	-5.6106E-02	-5.5316E-02
A6 =	-1.2366E-02	1.3448E-01	3.5562E-01	5.1022E-02	-6.3646E-02	-5.7367E-02
A8 =	3.4980E-02	-1.9148E-01	-5.3260E-01	2.3829E-02	1.7648E-01	1.7934E-01
A10 =	-6.5342E-02	2.0254E-01	5.3827E-01	-1.5043E-01	-2.7542E-01	-3.4955E-01
A12 =	5.1076E-02	-1.4376E-01	-3.0973E-01	1.8139E-01	2.2466E-01	3.7429E-01
A14 =	-1.8011E-02	3.9150E-02	7.3045E-02	-6.4338E-02	-6.1189E-02	-2.0194E-01
A16 =	6.3836E-05	-2.4284E-05	-4.8693E-05	4.2676E-06	1.0878E-05	4.6556E-02
Surface #	8	9	10	11	12	13
k =	-6.8399E-01	4.0834E+00	-3.2760E+01	-3.7700E+01	-2.5217E+00	-1.1626E+00
A4 =	1.3533E-03	3.2070E-02	4.2013E-02	-1.4141E-03	-1.4107E-01	-1.0722E-01
A6 =	4.3848E-02	-2.2431E-02	-1.1475E-01	-6.4857E-03	6.3037E-02	3.5611E-02
A8 =	-1.0208E-01	-1.8953E-02	7.5519E-02	1.1997E-04	-1.8397E-02	-9.6073E-03
A10 =	5.8869E-02	1.1540E-02	-4.5883E-02	3.5938E-04	3.4199E-03	1.7203E-03
A12 =	-1.4059E-02	-1.5302E-03	1.5590E-02	-5.2960E-05	-3.8001E-04	-1.8819E-04
A14 =	-4.1136E-04	5.2927E-06	-2.1431E-03	1.4337E-06	2.2951E-05	1.1208E-05
A16 =	2.2488E-11	1.4692E-06	9.8086E-05	6.1775E-08	-5.8161E-07	-2.7566E-07

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment		
f [mm]	4.83	(R9 + R10)/(R9 - R10)
Fno	2.20	f/R4
HFOV [deg.]	38.1	f1/f
V5	30.2	f1/f2
(R1 + R2)/(R1 - R2)	-1.00	ΣCT/Td
(R3 + R4)/(R3 - R4)	-1.11	TL/ImgH

4th Embodiment

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment.

In FIG. 7, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the

present disclosure and an image sensor 490. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image plane 480, wherein the photographing lens assembly has a total of six non-cemented lens elements (410-460) with refractive power.

The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being convex in a paraxial region thereof. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being both aspheric.

The second lens element 420 with negative refractive power has an object-side surface 421 being concave in a paraxial region thereof and an image-side surface 422 being convex in a paraxial region thereof. The second lens element 420 is made of plastic material and has the object-side surface 421 and the image-side surface 422 being both aspheric, wherein both of the object-side surface 421 and the image-side surface 422 of the second lens element 420 have at least one inflection point.

The third lens element 430 with negative refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof. The third lens element 430 is made of plastic material and has the object-side surface 431 and the image-side surface 432 being both aspheric, wherein both of the object-side surface 431 and the image-side surface 432 of the third lens element 430 have at least one inflection point.

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The fourth lens element **440** with negative refractive power has an object-side surface **441** being concave in a paraxial region thereof and an image-side surface **442** being concave in a paraxial region thereof. The fourth lens element **440** is made of plastic material and has the object-side surface **441** and the image-side surface **442** being both aspheric, wherein the image-side surface **442** of the fourth lens element **440** has at least one inflection point.

The fifth lens element **450** with negative refractive power has an object-side surface **451** being concave in a paraxial region thereof and an image-side surface **452** being concave in a paraxial region thereof. The fifth lens element **450** is made of plastic material and has the object-side surface **451** and the image-side surface **452** being both aspheric, wherein

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region thereof and an image-side surface **462** being concave in a paraxial region thereof, wherein the image-side surface **462** of the sixth lens element **460** has at least one convex shape in an off-axis region thereof. The sixth lens element **460** is made of plastic material and has the object-side surface **461** and the image-side surface **462** being both aspheric.

The IR-cut filter **470** is made of glass and located between the sixth lens element **460** and the image plane **480**, and will not affect the focal length of the photographing lens assembly. The image sensor **490** is disposed on the image plane **480** of the photographing lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment								
f = 5.22 mm, Fno = 2.35, HFOV = 36.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.304					
2	Lens 1	1.937 ASP	0.776	Plastic	1.544	55.9	3.20	
3		-14.933 ASP	0.060					
4	Lens 2	-4.199 ASP	0.280	Plastic	1.634	23.8	-7.99	
5		-25.196 ASP	0.358					
6	Lens 3	7.136 ASP	0.287	Plastic	1.544	55.9	-174.14	
7		6.542 ASP	0.377					
8	Lens 4	-216.942 ASP	0.326	Plastic	1.634	23.8	-174.11	
9		224.863 ASP	0.356					
10	Lens 5	-31.804 ASP	0.619	Plastic	1.607	26.6	-17.03	
11		15.434 ASP	0.177					
12	Lens 6	3.450 ASP	1.261	Plastic	1.535	55.7	-24.18	
13		2.376 ASP	0.300					
14	IR- cut filter	Plano	0.210	Glass	1.517	64.2	—	
15		Plano	0.586					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 7 is 1.110 mm.

TABLE 8

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-3.6210E-02	-9.0000E+01	-4.7175E+01	-6.8681E+01	-9.0000E+01	-1.0000E+00
A4 =	-2.4738E-03	-6.2001E-02	-9.0087E-02	2.2476E-02	-4.0809E-02	-5.8868E-02
A6 =	-1.0983E-02	8.1703E-02	2.2236E-01	3.0846E-02	-3.8188E-02	2.9160E-02
A8 =	1.6634E-02	-9.9086E-02	-2.7255E-01	1.2597E-02	9.1279E-02	-4.8148E-02
A10 =	-2.8096E-02	8.5500E-02	2.2867E-01	-6.3525E-02	-1.1648E-01	2.3891E-02
A12 =	1.7629E-02	-5.0606E-02	-1.0831E-01	6.3478E-02	7.8919E-02	1.1239E-02
A14 =	-5.6132E-03	1.1242E-02	2.1304E-02	-1.8899E-02	-1.7622E-02	-1.7569E-02
A16 =	-5.2751E-04	-2.0947E-05	5.8616E-05	-2.5705E-04	5.6251E-05	6.2733E-03
Surface #	8	9	10	11	12	13
k =	1.0000E+00	1.0000E+00	-3.2760E+01	-3.7700E+01	-1.1888E+01	-1.3938E+00
A4 =	-5.7355E-02	-4.0580E-02	4.4646E-02	-1.9163E-02	-1.0803E-01	-8.0849E-02
A6 =	3.1370E-02	-1.4675E-02	-9.4895E-02	7.7369E-03	3.9914E-02	2.2553E-02
A8 =	-5.2979E-02	8.5632E-03	5.9259E-02	-9.3279E-03	-9.4207E-03	-4.9856E-03
A10 =	2.2514E-02	-1.8917E-02	-3.4086E-02	3.8273E-03	1.4482E-03	7.2944E-04
A12 =	-5.8119E-03	1.8787E-02	1.1290E-02	-7.4676E-04	-1.3346E-04	-6.6068E-05
A14 =	-9.3102E-04	-9.6624E-03	-1.7159E-03	7.2225E-05	6.6202E-06	3.2430E-06
A16 =	-5.0454E-04	2.0587E-03	5.5278E-05	-2.8439E-06	-1.3565E-07	-6.4201E-08

the image-side surface **452** of the fifth lens element **450** has at least one convex shape in an off-axis region thereof.

The sixth lens element **460** with negative refractive power has an object-side surface **461** being convex in a paraxial

In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those

stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	5.22	$(R9 + R10)/(R9 - R10)$	0.35
Fno	2.35	$f/R4$	-0.21
HFOV [deg.]	36.0	$f1/f$	0.61
V5	26.6	$f1/f2$	0.40
$(R1 + R2)/(R1 - R2)$	-0.77	$\Sigma CT/Td$	0.73
$(R3 + R4)/(R3 - R4)$	-1.40	$TL/ImgH$	1.55

5th Embodiment

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment.

In FIG. 9, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 590. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 570 and an image plane 580, wherein the photographing lens assembly has a total of six non-cemented lens elements (510-560) with refractive power.

The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof. The first lens element 510 is made of plastic material and has the object-side surface 511 and the image-side surface 512 being both aspheric.

The second lens element 520 with positive refractive power has an object-side surface 521 being concave in a paraxial region thereof and an image-side surface 522 being convex in a paraxial region thereof. The second lens element 520 is made of plastic material and has the object-side surface 521

and the image-side surface 522 being both aspheric, wherein both of the object-side surface 521 and the image-side surface 522 of the second lens element 520 have at least one inflection point.

The third lens element 530 with positive refractive power has an object-side surface 531 being concave in a paraxial region thereof and an image-side surface 532 being convex in a paraxial region thereof. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being both aspheric, wherein both of the object-side surface 531 and the image-side surface 532 of the third lens element 530 have at least one inflection point.

The fourth lens element 540 with negative refractive power has an object-side surface 541 being concave in a paraxial region thereof and an image-side surface 542 being convex in a paraxial region thereof. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being both aspheric, wherein the image-side surface 542 of the fourth lens element 540 has at least one inflection point.

The fifth lens element 550 with negative refractive power has an object-side surface 551 being concave in a paraxial region thereof and an image-side surface 552 being concave in a paraxial region thereof. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being both aspheric, wherein the image-side surface 552 of the fifth lens element 550 has at least one convex shape in an off-axis region thereof.

The sixth lens element 560 with negative refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being concave in a paraxial region thereof, wherein the image-side surface 562 of the sixth lens element 560 has at least one convex shape in an off-axis region thereof. The sixth lens element 560 is made of plastic material and has the object-side surface 561 and the image-side surface 562 being both aspheric.

The IR-cut filter 570 is made of glass and located between the sixth lens element 560 and the image plane 580, and will not affect the focal length of the photographing lens assembly. The image sensor 590 is disposed on the image plane 580 of the photographing lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 5.16 mm, Fno = 2.45, HFOV = 36.1 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.270					
2	Lens 1	1.843 ASP	0.654	Plastic	1.544	55.9	3.66	
3		21.938 ASP	0.067					
4	Lens 2	-6.244 ASP	0.264	Plastic	1.634	23.8	2580.16	
5		-6.322 ASP	0.333					
6	Lens 3	-6.470 ASP	0.417	Plastic	1.544	55.9	51.19	
7		-5.369 ASP	0.337					
8	Lens 4	-2.705 ASP	0.793	Plastic	1.634	23.8	-52.37	
9		-3.280 ASP	0.086					
10	Lens 5	-23.915 ASP	1.237	Plastic	1.634	23.8	-5.70	
11		4.344 ASP	0.291					
12	Lens 6	2.535 ASP	0.652	Plastic	1.562	40.5	-75.92	
13		2.172 ASP	0.300					

TABLE 9-continued

5th Embodiment						
f = 5.16 mm, Fno = 2.45, HFOV = 36.1 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Focal Length
14	IR- cut filter	Plano	0.300	Glass	1.517	64.2
15		Plano	0.435			
16	Image	Plano	—			

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-5.6633E-02	-9.0000E+01	-9.0000E+01	-1.5445E+00	-9.0000E+01	1.6217E+00
A4 =	-6.4074E-03	-9.5242E-02	-1.2723E-01	2.5039E-02	-3.5096E-02	-1.8745E-02
A6 =	-1.6631E-02	1.2349E-01	3.5860E-01	4.6232E-02	-4.6862E-02	-5.3864E-04
A8 =	3.6746E-02	-1.7981E-01	-5.4375E-01	5.7505E-02	1.5646E-01	1.1584E-02
A10 =	-7.9371E-02	1.8557E-01	5.6825E-01	-2.0649E-01	-2.5149E-01	-1.8696E-01
A12 =	6.5299E-02	-1.3434E-01	-3.2931E-01	2.3227E-01	2.0402E-01	3.0035E-01
A14 =	-2.8101E-02	3.8545E-02	8.1796E-02	-8.7346E-02	-5.2034E-02	-1.9805E-01
A16 =	1.1684E-03	-1.3625E-03	-3.5588E-03	2.7859E-03	-8.5479E-04	5.1730E-02
Surface #	8	9	10	11	12	13
k =	9.9949E-01	-8.7407E+00	-3.2760E+01	-3.7907E+01	-4.3775E+00	-1.1869E+00
A4 =	-2.4891E-03	-4.0337E-02	-5.7833E-02	-5.4865E-02	-1.4697E-01	-1.0803E-01
A6 =	-1.0674E-02	4.7067E-02	4.1441E-02	4.1395E-02	6.6010E-02	3.5469E-02
A8 =	-9.2395E-03	-1.0749E-02	-5.4706E-03	-1.9523E-02	-1.9885E-02	-9.6268E-03
A10 =	-9.0033E-02	-4.8186E-02	-3.8212E-02	5.1597E-03	3.8880E-03	1.8034E-03
A12 =	9.8748E-02	4.4229E-02	3.0971E-02	-7.6704E-04	-4.5536E-04	-2.0793E-04
A14 =	-4.7253E-02	-1.5531E-02	-1.0092E-02	6.0633E-05	2.8983E-05	1.2964E-05
A16 =	7.2978E-03	2.1000E-03	1.2555E-03	-1.9989E-06	-7.7291E-07	-3.2981E-07

In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment		
f [mm]	5.16 (R9 + R10)/(R9 - R10)	0.69
Fno	2.45 f/R4	-0.82
HFOV [deg.]	36.1 f1/f	0.71
V5	23.8 f1/f2	0.001
(R1 + R2)/(R1 - R2)	-1.18 ΣCT/Td	0.78
(R3 + R4)/(R3 - R4)	-160.49 TL/ImgH	1.60

6th Embodiment

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment.

In FIG. 11, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 690. The photo-

graphing lens assembly includes, in order from an object side to an image side, a first lens element 610, an aperture stop 600, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image plane 680, wherein the photographing lens assembly has a total of six non-cemented lens elements (610-660) with refractive power.

The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being convex in a paraxial region thereof. The first lens element 610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

The second lens element 620 with negative refractive power has an object-side surface 621 being concave in a paraxial region thereof and an image-side surface 622 being convex in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric, wherein both of the object-side surface 621 and the image-side surface 622 of the second lens element 620 have at least one inflection point.

The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being concave in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric, wherein both of the object-side surface 631 and the image-side surface 632 of the third lens element 630 have at least one inflection point.

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The fourth lens element **640** with positive refractive power has an object-side surface **641** being concave in a paraxial region thereof and an image-side surface **642** being convex in a paraxial region thereof. The fourth lens element **640** is made of plastic material and has the object-side surface **641** and the image-side surface **642** being both aspheric, wherein the image-side surface **642** of the fourth lens element **640** has at least one inflection point. Furthermore, the positive refractive power of the fourth lens element **640** becomes weak from a paraxial region towards a peripheral region.

The fifth lens element **650** with negative refractive power has an object-side surface **651** being concave in a paraxial region thereof and an image-side surface **652** being concave in a paraxial region thereof. The fifth lens element **650** is

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The sixth lens element **660** with negative refractive power has an object-side surface **661** being convex in a paraxial region thereof and an image-side surface **662** being concave in a paraxial region thereof, wherein the image-side surface **662** of the sixth lens element **660** has at least one convex shape in an off-axis region thereof. The sixth lens element **660** is made of plastic material and has the object-side surface **661** and the image-side surface **662** being both aspheric.

The IR-cut filter **670** is made of glass and located between the sixth lens element **660** and the image plane **680**, and will not affect the focal length of the photographing lens assembly. The image sensor **690** is disposed on the image plane **680** of the photographing lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment							
f = 5.55 mm, Fno = 2.28, HFOV = 35.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	2.002 ASP	0.837	Plastic	1.544	55.9	3.55
2		-47.592 ASP	-0.041				
3	Ape. Stop	Plano	0.139				
4	Lens 2	-5.196 ASP	0.250	Plastic	1.634	23.8	-8.65
5		-100.000 ASP	0.288				
6	Lens 3	5.850 ASP	0.365	Plastic	1.544	55.9	18.91
7		13.262 ASP	0.436				
8	Lens 4	-4.279 ASP	0.752	Plastic	1.634	23.8	268.04
9		-4.459 ASP	0.222				
10	Lens 5	-101.272 ASP	0.869	Plastic	1.583	30.2	-7.51
11		4.594 ASP	0.277				
12	Lens 6	3.465 ASP	1.009	Plastic	1.535	55.7	-36.46
13		2.644 ASP	0.300				
14	IR-cut filter	Plano	0.175	Glass	1.517	64.2	—
15		Plano	0.544				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients						
Surface #	1	2	4	5	6	7
k =	-1.0001E+00	-9.0000E+01	-9.0000E+01	-9.0000E+01	-3.6912E+01	3.8823E-01
A4 =	8.6777E-03	-2.2733E-02	-6.1707E-02	1.7679E-02	-2.9398E-02	-1.5742E-02
A6 =	6.9754E-03	9.4701E-03	1.3636E-01	-8.7136E-03	5.2942E-03	-4.4720E-02
A8 =	-7.2223E-03	-4.1389E-02	-2.1748E-01	3.7975E-02	-9.6158E-02	3.3241E-02
A10 =	-7.3052E-03	9.0885E-02	2.9856E-01	-3.3205E-02	1.9316E-01	-4.4519E-02
A12 =	1.4893E-02	-1.0020E-01	-2.6410E-01	7.9529E-03	-1.8091E-01	5.4013E-02
A14 =	-1.0043E-02	5.2493E-02	1.2851E-01	4.5922E-03	8.4090E-02	-3.5063E-02
A16 =	2.0877E-03	-1.0545E-02	-2.5336E-02	-1.1971E-03	-1.4374E-02	8.7600E-03
Surface #	8	9	10	11	12	13
k =	2.5581E-01	-6.4759E+00	-3.2760E+01	-3.7700E+01	-1.3691E+00	-1.1739E+00
A4 =	1.8109E-02	4.7322E-02	2.2810E-02	-3.5218E-02	-1.3266E-01	-7.7262E-02
A6 =	-7.9950E-02	-8.8378E-02	-5.9943E-02	2.5381E-02	5.1058E-02	2.1143E-02
A8 =	6.4696E-02	6.8608E-02	3.2180E-02	-1.3627E-02	-1.2479E-02	-4.8257E-03
A10 =	-3.5318E-02	-4.4493E-02	-1.4906E-02	3.9581E-03	2.0818E-03	7.4383E-04
A12 =	-2.1465E-02	1.9282E-02	2.8033E-03	-6.4722E-04	-2.2703E-04	-6.9351E-05
A14 =	3.3589E-02	-4.5181E-03	1.4445E-04	5.7048E-05	1.4398E-05	3.5063E-06
A16 =	-1.2010E-02	4.6651E-04	-5.7190E-05	-2.1255E-06	-3.9957E-07	-7.4039E-08

made of plastic material and has the object-side surface **651** and the image-side surface **652** being both aspheric, wherein the image-side surface **652** of the fifth lens element **650** has at least one convex shape in an off-axis region thereof.

In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those

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stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	5.55	$(R9 + R10)/(R9 - R10)$	0.91
Fno	2.28	$f/R4$	-0.06
HFOV [deg.]	35.0	$f1/f$	0.64
V5	30.2	$f1/f2$	0.41
$(R1 + R2)/(R1 - R2)$	-0.92	$\Sigma CT/Td$	0.76
$(R3 + R4)/(R3 - R4)$	-1.11	$TL/ImgH$	1.66

7th Embodiment

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment.

In FIG. 13, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 790. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 700, a first lens element 710, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 770 and an image plane 780, wherein the photographing lens assembly has a total of six non-cemented lens elements (710-760) with refractive power.

The first lens element 710 with positive refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being convex in a paraxial region thereof. The first lens element 710 is made of plastic material and has the object-side surface 711 and the image-side surface 712 being both aspheric.

The second lens element 720 with negative refractive power has an object-side surface 721 being concave in a paraxial region thereof and an image-side surface 722 being concave in a paraxial region thereof. The second lens element 720 is made of plastic material and has the object-side surface

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721 and the image-side surface 722 being both aspheric, wherein the object-side surface 721 of the second lens element 720 has at least one inflection point.

The third lens element 730 with positive refractive power has an object-side surface 731 being convex in a paraxial region thereof and an image-side surface 732 being concave in a paraxial region thereof. The third lens element 730 is made of plastic material and has the object-side surface 731 and the image-side surface 732 being both aspheric, wherein both of the object-side surface 731 and the image-side surface 732 of the third lens element 730 have at least one inflection point.

The fourth lens element 740 with positive refractive power has an object-side surface 741 being concave in a paraxial region thereof and an image-side surface 742 being convex in a paraxial region thereof. The fourth lens element 740 is made of plastic material and has the object-side surface 741 and the image-side surface 742 being both aspheric. Furthermore, the positive refractive power of the fourth lens element 740 becomes weak from a paraxial region towards a peripheral region.

The fifth lens element 750 with negative refractive power has an object-side surface 751 being concave in a paraxial region thereof and an image-side surface 752 being concave in a paraxial region thereof. The fifth lens element 750 is made of plastic material and has the object-side surface 751 and the image-side surface 752 being both aspheric, wherein the image-side surface 752 of the fifth lens element 750 has at least one convex shape in an off-axis region thereof.

The sixth lens element 760 with negative refractive power has an object-side surface 761 being convex in a paraxial region thereof and an image-side surface 762 being concave in a paraxial region thereof, wherein the image-side surface 762 of the sixth lens element 760 has at least one convex shape in an off-axis region thereof. The sixth lens element 760 is made of plastic material and has the object-side surface 761 and the image-side surface 762 being both aspheric.

The IR-cut filter 770 is made of glass and located between the sixth lens element 760 and the image plane 780, and will not affect the focal length of the photographing lens assembly. The image sensor 790 is disposed on the image plane 780 of the photographing lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment							
f = 4.87 mm, Fno = 2.20, HFOV = 37.8 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.310				
2	Lens 1	1.883 ASP	0.900	Plastic	1.544	55.9	3.10
3		-13.691 ASP	0.051				
4	Lens 2	-4.373 ASP	0.250	Plastic	1.650	21.4	-6.04
5		39.361 ASP	0.232				
6	Lens 3	5.086 ASP	0.320	Plastic	1.544	55.9	36.94
7		6.658 ASP	0.327				
8	Lens 4	-11.974 ASP	0.300	Plastic	1.607	26.6	20.15
9		-6.110 ASP	0.243				
10	Lens 5	-45.120 ASP	0.760	Plastic	1.544	55.9	-8.45
11		5.153 ASP	0.262				
12	Lens 6	2.884 ASP	0.990	Plastic	1.535	55.7	-26.67
13		2.112 ASP	0.300				

TABLE 13-continued

7th Embodiment f = 4.87 mm, Fno = 2.20, HFOV = 37.8 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Focal Length
14	IR- cut filter	Plano	0.210	Glass	1.517	64.2
15		Plano	0.536			
16	Image	Plano	—			

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 12 is 2.800 mm.

TABLE 14

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-3.2438E-02	-1.0000E+00	-6.6366E+01	-1.0000E+00	-2.1886E+01	-1.0000E+00
A4 =	-6.2195E-03	-7.3281E-02	-1.0905E-01	2.0799E-02	-6.3093E-02	-5.3398E-02
A6 =	-1.3335E-02	1.3322E-01	3.5754E-01	5.2065E-02	-6.4467E-02	2.5324E-02
A8 =	3.5457E-02	-1.9143E-01	-5.3415E-01	2.4438E-02	1.7759E-01	-1.6100E-02
A10 =	-6.6414E-02	2.0456E-01	5.3813E-01	-1.5204E-01	-2.7425E-01	-1.6767E-01
A12 =	5.0415E-02	-1.4194E-01	-3.0868E-01	1.7926E-01	2.2545E-01	3.2854E-01
A14 =	-1.8547E-02	3.9229E-02	7.3675E-02	-6.5060E-02	-6.0702E-02	-2.3545E-01
A16 =	1.8110E-03	-5.4533E-04	-1.0706E-03	1.9299E-03	6.8547E-05	6.3236E-02
Surface #	8	9	10	11	12	13
k =	-1.1582E+00	-6.7431E+00	-3.2760E+01	-3.7621E+01	-1.8133E+00	-1.1280E+00
A4 =	-1.6080E-03	1.3000E-01	1.4526E-01	3.4965E-02	-1.3388E-01	-1.1126E-01
A6 =	2.8309E-01	-1.7086E-01	-2.9868E-01	-3.5036E-02	5.8881E-02	3.7732E-02
A8 =	-9.4995E-01	5.9361E-02	2.7178E-01	1.3540E-02	-1.6705E-02	-9.9372E-03
A10 =	1.4777E+00	4.8313E-02	-1.8089E-01	-3.5095E-03	2.9894E-03	1.6638E-03
A12 =	-1.2754E+00	-5.8002E-02	7.4614E-02	6.0300E-04	-3.1700E-04	-1.6431E-04
A14 =	5.7667E-01	2.1456E-02	-1.6386E-02	-5.8112E-05	1.8081E-05	8.6248E-06
A16 =	-1.0829E-01	-2.7278E-03	1.4919E-03	2.2646E-06	-4.2558E-07	-1.8189E-07

In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment		
f [mm]	4.87	(R9 + R10)/(R9 - R10)
Fno	2.20	f/R4
HFOV [deg.]	37.8	f1/f
V5	55.9	f1/ f2
(R1 + R2)/(R1 - R2)	-0.76	ΣCT/Td
(R3 + R4)/(R3 - R4)	-0.80	TL/ImgH

8th Embodiment

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment.

In FIG. 15, the image capturing device includes the photographing lens assembly (not otherwise herein labeled) of

the present disclosure and an image sensor 890. The photographing lens assembly includes, in order from an object side to an image side, an aperture stop 800, a first lens element 810, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, an IR-cut filter 870 and an image plane 880, wherein the photographing lens assembly has a total of six non-cemented lens elements (810-860) with refractive power.

The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being convex in a paraxial region thereof. The first lens element 810 is made of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric.

The second lens element 820 with negative refractive power has an object-side surface 821 being concave in a paraxial region thereof and an image-side surface 822 being convex in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric, wherein both of the object-side surface 821 and the image-side surface 822 of the second lens element 820 have at least one inflection point.

The third lens element 830 with negative refractive power has an object-side surface 831 being concave in a paraxial region thereof and an image-side surface 832 being concave in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric, wherein the image-side surface 832 of the third lens element 830 has at least one inflection point.

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The fourth lens element **840** with positive refractive power has an object-side surface **841** being concave in a paraxial region thereof and an image-side surface **842** being convex in a paraxial region thereof. The fourth lens element **840** is made of plastic material and has the object-side surface **841** and the image-side surface **842** being both aspheric, wherein both of the object-side surface **841** and the image-side surface **842** of the fourth lens element **840** have at least one inflection point. Furthermore, the positive refractive power of the fourth lens element **840** becomes weak from a paraxial region towards a peripheral region.

The fifth lens element **850** with negative refractive power has an object-side surface **851** being concave in a paraxial region thereof and an image-side surface **852** being concave in a paraxial region thereof. The fifth lens element **850** is made of plastic material and has the object-side surface **851**

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The sixth lens element **860** with positive refractive power has an object-side surface **861** being convex in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof, wherein the image-side surface **862** of the sixth lens element **860** has at least one convex shape in an off-axis region thereof. The sixth lens element **860** is made of plastic material and has the object-side surface **861** and the image-side surface **862** being both aspheric.

The IR-cut filter **870** is made of glass and located between the sixth lens element **860** and the image plane **880**, and will not affect the focal length of the photographing lens assembly. The image sensor **890** is disposed on the image plane **880** of the photographing lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15, and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment							
f = 2.86 mm, Fno = 2.05, HFOV = 38.4 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.152				
2	Lens 1	1.575 ASP	0.385	Plastic	1.544	55.9	2.49
3		-8.872 ASP	0.085				
4	Lens 2	-3.486 ASP	0.230	Plastic	1.634	23.8	-6.56
5		-22.103 ASP	0.419				
6	Lens 3	-49.567 ASP	0.284	Plastic	1.634	23.8	-15.92
7		12.706 ASP	0.214				
8	Lens 4	-3.343 ASP	0.750	Plastic	1.544	55.9	1.48
9		-0.702 ASP	0.231				
10	Lens 5	-82.275 ASP	0.230	Plastic	1.544	55.9	-1.55
11		0.851 ASP	0.108				
12	Lens 6	2.591 ASP	0.331	Plastic	1.614	25.6	2673.99
13		2.469 ASP	0.360				
14	IR- cut filter	Plano	0.210	Glass	1.517	64.2	—
15		Plano	0.230				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-5.7206E+00	-2.7249E+01	-7.5698E+01	-1.0000E+00	-1.0000E+00	8.9948E+01
A4 =	1.6239E-01	3.8940E-02	-5.7007E-02	1.3079E-01	-1.7237E-01	-6.4051E-02
A6 =	-2.5585E-02	1.0705E-01	5.8275E-01	-4.0948E-01	-8.2379E-01	-5.0532E-01
A8 =	-3.2867E-01	-8.3740E-01	-2.0802E+00	1.1604E+00	3.2188E+00	1.2578E+00
A10 =	8.1576E-01	2.2233E+00	4.2624E+00	-2.3904E+00	-7.5506E+00	-1.8871E+00
A12 =	-1.0378E+00	-3.0022E+00	-4.9158E+00	2.8601E+00	1.0185E+01	1.9611E+00
A14 =	6.4538E-01	1.8875E+00	2.8855E+00	-1.8838E+00	-7.0956E+00	-1.1645E+00
A16 =	-1.8782E-01	-4.7943E-01	-6.3561E-01	5.0695E-01	1.9743E+00	2.9378E-01
Surface #	8	9	10	11	12	13
k =	4.6978E+00	-3.8217E+00	-1.0000E+00	-6.7360E+00	-3.0024E-01	-5.6415E-01
A4 =	5.7434E-02	-2.7109E-01	-2.8372E-01	-9.5289E-01	-5.0917E-01	1.0900E-01
A6 =	1.0856E-01	3.6881E-01	5.2253E-01	1.6405E+00	7.5366E-01	-2.3635E-01
A8 =	-9.3685E-01	-3.6595E-01	-6.2559E-01	-1.4066E+00	-6.1328E-01	1.5146E-01
A10 =	1.7347E+00	-2.2306E-02	4.2495E-01	6.4370E-01	2.5553E-01	-5.1910E-02
A12 =	-1.3661E+00	3.0745E-01	-1.5966E-01	-1.6139E-01	-5.5674E-02	9.8357E-03
A14 =	4.9719E-01	-1.8093E-01	3.1159E-02	2.0908E-02	5.9893E-03	-9.5868E-04
A16 =	-6.7591E-02	3.1331E-02	-2.4739E-03	-1.0933E-03	-2.4900E-04	3.6925E-05

and the image-side surface **852** being both aspheric, wherein the image-side surface **852** of the fifth lens element **850** has at least one convex shape in an off-axis region thereof.

In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these

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parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment		
f [mm]	2.86	$(R9 + R10)/(R9 - R10)$ 0.98
Fno	2.05	$f/R4$ -0.13
HFOV [deg.]	38.4	$f1/f$ 0.87
V5	55.9	$f1/f2$ 0.38
$(R1 + R2)/(R1 - R2)$	-0.70	$\Sigma CT/Td$ 0.68
$(R3 + R4)/(R3 - R4)$	-1.37	$TL/ImgH$ 1.77

9th Embodiment

FIG. 17 is a schematic view of a mobile terminal 10 according to the 9th embodiment of the present disclosure. The mobile terminal 10 of the 9th embodiment is a smart phone, wherein the mobile terminal 10 includes an image capturing device 11. The image capturing device 11 includes a photographing lens assembly (not otherwise herein labeled) according to the present disclosure and an image sensor (not otherwise herein labeled), wherein the image sensor is disposed on an image plane of the photographing lens assembly.

10th Embodiment

FIG. 18 is a schematic view of a mobile terminal 20 according to the 10th embodiment of the present disclosure. The mobile terminal 20 of the 10th embodiment is a tablet personal computer, wherein the mobile terminal 20 includes an image capturing device 21. The image capturing device 21 includes a photographing lens assembly (not otherwise herein labeled) according to the present disclosure and an image sensor (not otherwise herein labeled), wherein the image sensor is disposed on an image plane of the photographing lens assembly.

11th Embodiment

FIG. 19 is a schematic view of a mobile terminal 30 according to the 11th embodiment of the present disclosure. The mobile terminal 30 of the 11th embodiment is a head-mounted display, wherein the mobile terminal 30 includes an image capturing device 31. The image capturing device 31 includes a photographing lens assembly (not otherwise herein labeled) according to the present disclosure and an image sensor (not otherwise herein labeled), wherein the image sensor is disposed on an image plane of the photographing lens assembly.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-16 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the

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scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A photographing lens assembly comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element;

a third lens element;

a fourth lens element;

a fifth lens element with negative refractive power having an object-side surface being concave in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof; and

a sixth lens element having an object-side surface being convex in a paraxial region thereof and a image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and the object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the photographing lens assembly has a total of six non-cemented lens elements, a focal length of the photographing lens assembly is f , a focal length of the first lens element is $f1$, a curvature radius of the object-side surface of the second lens element is $R3$, a curvature radius of an image-side surface of the second lens element is $R4$, and the following conditions are satisfied:

$$0 < f1/f < 2.5;$$

$$(R3+R4)/(R3-R4) < -0.40; \text{ and}$$

$$-2.0 < f/R4 \leq 0.0.$$

2. The photographing lens assembly of claim 1, wherein the fourth lens element has an image-side surface being convex in a paraxial region thereof.

3. The photographing lens assembly of claim 1, wherein the second lens element has negative refractive power.

4. The photographing lens assembly of claim 1, wherein the focal length of the photographing lens assembly is f , the focal length of the first lens element is $f1$, and the following condition is satisfied:

$$0.30 < f1/f < 1.25.$$

5. The photographing lens assembly of claim 1, wherein an Abbe number of the fifth lens element is $V5$, and the following condition is satisfied:

$$V5 < 32.$$

6. The photographing lens assembly of claim 1, wherein the fourth lens element has an object-side surface being concave in a paraxial region thereof.

7. The photographing lens assembly of claim 1, wherein a curvature radius of the object-side surface of the fifth lens element is $R9$, a curvature radius of the image-side surface of the fifth lens element is $R10$, and the following condition is satisfied:

$$0 < (R9+R10)/(R9-R10) < 1.0.$$

8. The photographing lens assembly of claim 1, wherein the fourth lens element has positive refractive power, and the positive refractive power of the fourth lens element becomes weak from a paraxial region towards a peripheral region.

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9. The photographing lens assembly of claim 1, wherein the focal length of the first lens element is $f1$, a focal length of the second lens element is $f2$, and the following condition is satisfied:

$$f1/f2 < 0.75.$$

10. The photographing lens assembly of claim 1, wherein each of the second lens element, the third lens element and the fourth lens element has at least one inflection point on at least one of the object-side surface and the image-side surface thereof.

11. The photographing lens assembly of claim 1, wherein both of the object-side surface and the image-side surface of the fifth lens element are aspheric and the image-side surface of the fifth lens element has at least one convex shape in an off-axis region thereof.

12. The photographing lens assembly of claim 1, wherein an axial distance between the object-side surface of the first lens element and an image plane is TL , a maximum image height of the photographing lens assembly is $ImgH$, and the following condition is satisfied:

$$TL/ImgH < 2.0.$$

13. The photographing lens assembly of claim 1, wherein a curvature radius of the object-side surface of the first lens element is $R1$, a curvature radius of an image-side surface of

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the first lens element is $R2$, and the following condition is satisfied:

$$-1.5 < (R1+R2)/(R1-R2) < -0.4.$$

14. The photographing lens assembly of claim 1, wherein the curvature radius of the object-side surface of the second lens element is $R3$, the curvature radius of the image-side surface of the second lens element is $R4$, and the following condition is satisfied:

$$-2.0 < (R3+R4)/(R3-R4) < -0.60.$$

15. The photographing lens assembly of claim 1, wherein a sum of the central thicknesses of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element is ΣCT , an axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is Td , and the following condition is satisfied:

$$0.65 < \Sigma CT/Td < 0.85.$$

16. An image capturing device, comprising:
the photographing lens assembly of claim 1; and
an image sensor, wherein the image sensor is located on an image plane of the photographing lens assembly.

17. A mobile terminal, comprising:
the image capturing device of claim 16.

18. The photographing lens assembly of claim 1, wherein the second lens element has an object-side surface being concave in a paraxial region thereof.

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